

Multiplicative Personalized PageRank

**A graph-based method for measuring information
hazards from dual-use biotechnology research**

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Abstract

Information hazards arise when the dissemination of true information enables harm. These risks are especially acute in dual-use emerging technologies such as biotechnology. A subset of information hazard, called attention hazards, occurs when drawing attention to an existing idea increases risks, like dual-use potential of a technology. Although widely acknowledged, current discussions around mitigating these hazards remain largely qualitative, with no systematic, data-driven approaches to measure attention hazards.

This dissertation develops a method to measure biotechnology attention hazards by examining five research papers of dual-use concern. A directed, ego-centric, heterogeneous citation network is constructed around these core papers, incorporating both academic and non-academic sources such as policy documents, news, blogs, and websites. To analyze this network, a new model called Multiplicative Personalized PageRank (MPPR) is introduced. MPPR extends traditional PageRank by integrating structural and temporal biases - weighting nodes by proximity to core papers and their recency - and by applying heterogeneous edge weights to reflect the amplification potential of different source types. MPPR scores (the share of probability mass sources accumulate in the network) serve as a proxy for attention, indicating each source's capacity to propagate information from core papers, thereby drawing attention to dual-use knowledge.

The findings show that MPPR functions as a middle ground approach, balancing the effects of more extreme biases on node distributions seen with alternative PageRank variants. Experiments demonstrate that MPPR shifts probability mass toward information hazard-related sources, particularly highly visible types such as news, blogs, policy documents, and videos; although gains are magnified by their small baseline presence. Edge weights exerted minimal influence on these outcomes, with structural position and recency of sources driving most effects. Overall, MPPR demonstrates a robust method for quantifying biotechnology attention hazards, offering a flexible platform that can be extended with refined node classification and edge weighting methods to better nuance the model.

Declaration

I have read and understood the College and Departmental statements and guidelines concerning plagiarism. I declare that:

- This submission is entirely my own original work.
- Wherever published, unpublished, printed, electronic or other information sources have been used as a contribution or component of this work, these are explicitly, clearly and individually acknowledged by appropriate use of quotation marks, citations, references and statements in the text. It is 14100 words in length.
- Any use of generative AI has been clearly acknowledged in my work and is permitted under the assignment's categorisation according to UCL's [AI guidelines](#).

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List of abbreviations

AI – artificial intelligence

AIHPR – average information hazard-related node percentile rank

API – Application Programming Interface

APR – average percentile rank

BW – base weight edge-weighting scheme

BW_TD – base weight $\times$ time-decay edge-weighting scheme

BW_TD_L – BW_TD $\times$ layer score edge-weighting scheme

BW_TD_LT – BW_TD_L $\times$ additional weights for cited nodes types edge-weighting scheme

CSV – comma-separated values

DOI – Digital Object Identifier

IH – information hazard-related

MPPR – Multiplicative Personalized PageRank

MWEs – multi-word expressions

NLP – natural language processing

Non-IH – non-information hazard-related

PPR – Personalized PageRank

Rev+MBV – weighted graph, reversed edges (citing $\rightarrow$ cited), MPPR bias vector

RWBC – random-walk betweenness centrality

SIR – the susceptible-infective-recovered (model)

SIS – susceptible-infective-susceptible (model)

UNW+MBV – unweighted graph, MPPR bias vector

UNW+UBV – unweighted graph, uniform bias vector

UUW+UBV – undirected, unweighted graph, uniform bias vector

VDA - Vargha and Delaney's A

W+DBV – weighted graph, bias restricted to time decay

W+LBV – weighted graph, bias restricted to layer scores

W+MWBV_{mw value} – weighted graph, mixed bias vector (layer–time decay combination, mw controls layer weight)

W+RBV – weighted graph, random bias vector

W+TBV – weighted graph, bias restricted to node type

W+UBV – weighted graph, uniform bias vector

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1 Introduction

Information hazards are ‘risks that arise from the dissemination or the potential dissemination of true information that may cause harm or enable some agent to cause harm’ [1, p.44]. A classic example is the risk of dissemination of classified information to adversarial state actors or non-state actors with malign intentions. Emerging technologies – characterized by radical novelty, relatively fast growth, coherence, prominent impact, uncertainty, and ambiguity¹ – create information through scientific and technological advances with the capacity to cause large-scale harm and potentially existential catastrophe if misused.²

Dual-use technologies, as technologies with both civilian and military applications that could be misused for malicious purposes beyond their original design [5, 6], are particularly prone to this type of risk. Biotechnology and artificial intelligence (AI) are two well cited examples in this regard. Synthetic biology, a subset of biotechnology, can already be used to engineer pathogens capable of causing pandemics; while a pathogen’s genetic information can be used to develop countermeasures such as vaccines, malicious actors could use the same information to increase a pathogen’s contagiousness to develop a biological weapon [7]. Additionally, there are concerns over increasing dual-use biotechnology risks through cheaper and easier access to software, hardware, databases, and information necessary to conduct DNA synthesis and gene editing through open source platforms [8, 9].

AI can amplify biotechnology dual-use risks. Open source generative AI models, including large language models, could help facilitate finding new chemical and

¹ Here I refer to an emerging technology as defined by Rotolo et al., [2, p.1830]: ‘a radically novel and relatively fast growing technology characterized by a certain degree of coherence persisting over time and with the potential to exert a considerable impact on the socio-economic domain(s) which is observed in terms of the composition of actors, institutions and patterns of interactions among those, along with the associated knowledge production processes. Its most prominent impact, however, lies in the future and so in the emergence phase is still somewhat uncertain and ambiguous.’

² Ord [3, p.37] defines existential catastrophe as ‘the destruction of humanity's long-term potential’, the outcome of which this potential is permanently lost. Examples of existential catastrophe include extinction events, unrecoverable collapse of society and unrecoverable dystopian situations. Bostrom [4, p.15] first introduced a related concept, existential risk, which is a risk ‘...that threatens the premature extinction of Earth-originating intelligent life or the permanent and drastic destruction of its potential for desirable future development’.

biological means for weapons [10]. Researchers have already shown how AI models used to assist in drug discovery can easily be converted to find and design new biochemical weapons [11]. These concerns have motivated some AI researchers to restrict certain information in their published research, such as model weights [10].

Attention hazards, defined as when ‘the mere drawing of attention to some particularly potent or relevant ideas or data increases risk, even when these ideas or data are already ‘known’’ [1, p.3], represent a subset of information hazard. Attention hazards are particularly salient in the domain of dual-use biotechnology, where legitimate scientific advances may simultaneously lower barriers for malicious uses. Despite recognition of these risks, there remains a lack of systematic, data-driven approaches to measure attention hazards and how they spread. Regulatory and scholarly discussions of information hazards from dual-use research of concern remain largely confined to qualitative policy proposals with no attempt to quantify information hazards [12, 13]. Without a way to quantify information hazards, decision-makers cannot rigorously evaluate the effectiveness of interventions or anticipate the reach of dual-use knowledge.

This study addresses that gap by developing a model to quantify the dissemination of attention hazards from five focal biotechnology research papers that have attracted wide concerns of dual-use applications [14-18]. To do so, a directed, ego-centric, heterogeneous citation network centered on these core papers is constructed. Unlike traditional bibliometric studies that focus narrowly on academic publications, this network explicitly incorporates diverse downstream sources - including policy documents, news media, blogs, and other websites – to capture attention across domains. The built network traces not only first-order sources (those directly citing core papers) but also second-order amplifiers (sources citing those first-order sources).

The methodological contribution of this dissertation lies in the adaptation of Personalized PageRank (PPR) models into a novel *Multiplicative Personalized PageRank* (MPPR) variant. MPPR incorporates:

1. Structural and temporal biases into a modifiable bias vector, influencing probability mass distribution over nodes. A *layer score* ensures that nodes

structurally closer to core papers receive more weight, while a *time decay* component reflects the empirical reality that attention to sources (through citations) generally diminishes with time.

2. Source-specific bias into edge weights to represent how information diffuses through heterogeneous networks according to the amount of attention sources attract. Edge weights are adapted from *Altmetric.com*-style attention scores [19] which capture the visibility and reach of different source types. For instance, a citation from a major news outlet is likely to amplify attention more strongly than a citation from a niche academic paper.

In this dissertation, MPPR scores are used as a proxy for attention, representing the stationary probability of residing at a given node in the network. To capture how information spreads, the edge orientation is inverted (cited → citing), so probability mass diffuses outward from core papers into their citing nodes and beyond. In this setup, a node's MPPR score reflects its capacity to amplify information from core nodes, providing a formal mechanism for measuring attention hazards: the more probability mass a node accumulates, the greater its role in propagating information from core papers of dual-use concern and in drawing attention to them.

To identify which sources engage with biotechnology information hazards and dual-use concerns, this study employs a keyword and key phrase-based lookup scoring method. Each node is assigned a score based on the presence of information hazard-related terms in its metadata (e.g., titles, summaries, assigned topic words), with a tiered weighting scheme distinguishing between critical terms (e.g., “bioterrorism”), high-relevance terms (e.g., “synthetic biology”), medium-relevance terms (e.g., “risk assessment”), and broad terms of potential interest (e.g., “regulation”). The result is a continuous score indicating a node's proximity to information hazard discourse, with a threshold score used to classify nodes as information hazard-related. For brevity, future references to things that are information hazard-related will be prefixed by ‘IH’, while those that are not will be prefixed with ‘non-IH’.

This study assesses MPPR's capacity measure attention hazards of dual-use biotechnology research against other PPR model variants in the following ways:

- A main comparative analysis evaluates how different models surface IH versus non-IH sources in top-k ranked nodes ($k = 100, 500, 1\%, 5\%$) and in average percentile ranks. Their effects on PPR score distributions and percentile density profiles are also visualized.
- Experiment 1, which assesses:
 - a. the impact of edge weights on MPPR's performance against a baseline variant.
 - b. the sensitivity of edge-weighting schemes to changes in α , a parameter dictating whether probability mass is redistributed along outgoing edges or redistributed across all nodes.
- Experiment 2, which investigates:
 - a. how IH node representation among the top-k ranked nodes changes under different weighted PPR variants with the same edge direction, compared to MPPR.
 - b. how MPPR shifts the distribution of probability mass by node types (policy papers, blogs etc.) and IH node status, compared to a baseline.
- Experiment 3, which conducts a correlation analysis to examine which variables (mean of outgoing edge weights, layer scores, and time decay) are most correlated with changes in PPR scores for IH nodes between a baseline and MPPR.

The main comparative analysis reveals that MPPR acts as a balanced middle ground approach among PPR variants, surfacing IH nodes in top-k rankings without systematic bias. Its layer score and time decay biases produce a wider spread of scores, accentuating contrasts between nodes without collapsing probability mass onto outliers or compressing distributions. Edge weights are found to have minimal impact on ranking outcomes across all variants.

Experiments 1a and 1b support the main comparative analysis, demonstrating that edge weight modifications have minimal impact on performance. Across base edge weights, and adding additional feature information, exponents, and varying α values to

edge weights, ranking changes are marginal. IH representation and average percentile ranks remain stable, with only slight decreases in top-K node overlap. This limited effect likely reflects the ego-centric structure of the network, where paths are restricted to two layers around five core papers, so topology dominates the flow of probability mass.

Experiment 2a confirms MPPR's role as a middle ground variant. The share of IH nodes across top-K subsets when averaged across variants is close to MPPR values, as variants that over and under-prioritize IH nodes offset each other. For Experiment 2b, MPPR redistributes probability mass share most toward IH nodes, particularly in high-attention source types such as blogs, Wikipedia articles, news media, videos, policy documents, books, and reports. These gains suggest IH nodes share the recency and proximity biases embedded in MPPR's bias vector. However, the large relative probability mass increase for most IH subgroups (e.g., +1613% for books) is partly an artefact of their small baseline shares compared to much larger non-IH groups, so the scale of changes should be interpreted with caution.

Experiment 3 confirms that MPPR's impact on IH nodes is driven primarily by structural position and recency rather than edge weights. Base edge weights show only weak associations with PPR scores changes, while newer nodes, and especially those located closer to core papers, gain disproportionately under MPPR.

This dissertation demonstrates that MPPR provides a useful foundation for quantifying biotechnology attention hazards. However, it does not explain why particular sources are influential, nor how external factors may shape the risk profile of a source, such as the threshold of technical knowledge required for malicious use. MPPR should be seen as a flexible model that can be extended, such as with Natural Language Processing (NLP)-based methods for IH source identification, and refined edge-weighting schemes.

2 Literature review

2.1 Information hazards and dual-use research

Despite the recognized risks of information hazards from dual-use research of concern, regulatory and academic discussions of information hazards remain largely qualitative, focusing on policy proposals [12, 13]. Existing work on information hazards is limited to classifying different types of information hazards [1], or applying these classifications in broader qualitative analyses [20, 21]. For dual-use biotechnology applications, mitigation strategies have focused on: screening requests for genetic materials; regulating synthetic DNA synthesis providers; and encouraging redaction or limited publication of sensitive research methodologies during peer review [22-24]. For AI dual-use concerns, mitigations include: restricting open-sourcing of powerful (generative) models [20]; developing frameworks for release of potential dual-use AI research [25, 26]; evaluating and red teaming AI models to identify vulnerabilities and potential for misuse [27-29]; studying potential motivations for misuse by malicious actors [21]; and assessing the potential future dual-use risks posed by generative models like large language models [30, 31].

What these approaches fail to offer is a quantifiable way to measure information hazards posed by specific publications or innovations. This highlights a critical gap in the literature: the absence of empirical, data-driven methods for assessing the dissemination and amplification of particular dual-use research.

2.2 Measuring knowledge diffusion in networks

Citation and social networks (also known as graphs) provide a well-established basis for studying knowledge diffusion. A social network is a ‘set of actors, or other entities, and a set or sets of relations defined on them’ [32, p.1], for instance a group of people who work at an organization, or people who follow each other on social media. Citation networks most often capture publication-publication, author-author, and author-publication links in an academic context.

Researchers are interested in how information or ideas travel within a network. Examples include the spread of ideas on social media between users [33, 34], the evolution of topics within a citation network [35], or even the diffusion of nanotechnology research between researchers [36]. The spread of information throughout a network depends not only on the information's intrinsic qualities, but also on the topology of the network [37].

Unlike existing qualitative approaches, citation network analysis offers a means to quantitatively measure information hazards. By modeling information flows from dual-use research papers to downstream amplifying heterogeneous sources, these links provide the means of tracing information hazards across not only academic, but policy and public discourse as well.

2.3 Approaches to network analysis

In the absence of direct measures of knowledge transmission, citations are commonly used as proxies for knowledge diffusion in scientific communities [38]. A directed citation graph can be represented as a directed graph $G = (V, E)$, where:

- $V = \{v_1, v_2, \dots, v_n\}$ is the set of nodes (also called vertices), with each node v_i representing a document (e.g., a paper, report, or policy document).
- $E \subseteq V \times V$ is the set of directed edges, where an edge $(v_i, v_j) \in E$ indicates that source v_i cites source v_j . In a directed graph, the direction of the edge encodes the flow of citation and therefore information.

2.3.1 Structural centrality measures

Networks are commonly analyzed at a global level to understand whole-of-network structures and characteristics [39]. These networks can be quite large, up to millions of nodes, even spanning entire disciplines or research fields [40]. Analyses of large-scale networks often apply centrality measures to identify influential papers, authors, or journals. For example [adapted from 39]:

- Degree centrality (C_D) measures the number of direct connections a node has, with high-degree nodes interpreted as having a more central role in the network

and greater ability to influence other nodes and spread information. Where $d(v_i)$ is the degree of node v_i :

$$C_D(v_i) = d(v_i) \quad (1)$$

- Betweenness centrality C_B quantifies how often a node appears on the shortest paths between other pairs of nodes in the network. It identifies nodes that act as bridges between different communities or subnetworks, and consequently have relatively stronger control over information flow. For equation 2 below, g_{jik} is all shortest paths linking node j and node k which pass through node i . g_{jk} is the shortest distance between nodes j and k :

$$C_B(v_i) = \sum_{j,k \neq i} \frac{g_{jik}}{g_{jk}} \quad (2)$$

These centrality measures are fundamentally structural, relying on network topology rather than information flow dynamics. Random-walk betweenness centrality (RWBC) relaxes the assumption that information spreads only along those shortest paths, by estimating how often a node is traversed by random walks between all node pairs in a network [41]. Though still structurally grounded, RWBC adopts a probabilistic view that captures information diffusion through multiple indirect paths as well as shortest routes.

2.3.2 Link-based centrality measures

Research has shifted toward link-based centrality measures, which account not only for a node's connections but also for the importance of its neighbors, offering a more nuanced view of how influence and information spread through networks. A seminal example is PageRank, a variant of eigenvector centrality originally developed by Brin and Page in 1998 [42] to rank web pages and which is now widely applied in bibliometrics, scientometrics, and network science. Emphasis on PageRank and its variants is deliberate here as they underpin MPPR.

PageRank builds on the intuition that a node is important if it is linked to by other important nodes, assigning each node a probability score through an iterative process that represents the long-run likelihood of reaching it via a random walk through the network. The algorithm (equation 3 below, from [43]) incorporates a damping factor $\alpha \in (0,1)$, modeling the probability that a random walk follows an outgoing edge versus teleporting to a random node. The damping factor balances two processes: with probability α , probability mass is redistributed along outgoing edges; with probability $(1 - \alpha)$ it is redistributed uniformly across all nodes. This mechanism guarantees convergence and prevents sinks (nodes with no outgoing edges) from absorbing all probability mass, distinguishing PageRank from other versions of eigenvector centrality that suffer from probability mass sinks and slow convergence [44].

At the outset, each node is initialized with an equal PageRank value, corresponding to a uniform probability distribution $\left(\frac{1}{N}\right)$ across the network. During each iteration, a node distributes its PageRank value evenly among all of its outgoing links to its neighbors. This process continues until the values converge to a steady state, meaning successive updates no longer produce significant changes in these values [44].

This recursive weighting process reflects the idea that influence is not distributed uniformly across a network, but accumulates through chains of endorsement, where authority begets further authority. In doing so, PageRank models more realistic diffusion of information and influence, where a node's visibility depends not only on its immediate popularity but also on its position within a broader network of significant actors. For the below equation:

$$r(p) = \alpha \sum_{q:(q,p) \in \mathcal{E}} \frac{r(q)}{\omega(q)} + (1 - \alpha) \frac{1}{N} \quad (3)$$

- $r(p)$ is the PageRank score of node p .
- The score of p is updated by summing contributions from all incoming neighbors q . Each node in q contributes a share of its own score $r(q)$ divided by its out-degree $\omega(q)$, scaled by the damping factor α .

- The term $\frac{1}{N}$, weighted by $(1 - \alpha)$, uniformly distributes a small baseline probability across all N nodes, ensuring all nodes receive some probability mass and that the iterative process converges.

In matrix equation form, PageRank operates as [adapted from 43]:

$$\mathbf{r}^{(k+1)} = \alpha \mathbf{T} \mathbf{r}^{(k)} + (1 - \alpha) \frac{1}{N} \mathbf{1}_N \quad (4)$$

Where:

- $\mathbf{r}$ is the PageRank vector, containing the score for every node.
 - $\mathbf{r}^{(k)}$ is the estimate of PageRank scores after k iterations.
 - $\mathbf{r}^{(k+1)}$ is the updated estimate at the next iteration.
 - When $\mathbf{r}^{(k+1)} = \mathbf{r}^{(k)}$ (within a tolerance) the process has converged to the stationary distribution.
- $\mathbf{T}$ is the transition probability matrix, encoding the entire link structure of the network. Each entry T_{pq} defines the probability of moving from node q to node p in a single step.
- $\mathbf{1}_N$ is a vector of ones of length N used to ensure every node has a nonzero baseline probability mass.

In citation network research, PageRank has been widely used to identify influential papers and journals beyond raw citation counts, offering a more nuanced view of scientific influence [45]. For scholarly knowledge diffusion, it has been applied to measure the diffusion of ideas within disciplines [35], identify high-impact and influential authors [46, 47], and measure academic institutional prestige [48]. Beyond bibliometrics, it has been applied in fields like biology for protein function prediction [49], and chemistry to analyze solvent structures [50].

PageRank's versatility stems from its capacity to be adapted, or 'personalized,' to suit different research contexts. This is achieved by introducing bias into the uniform probability mass vector and/or transition probability matrix, depending on the

researcher's aims [51]. In this way, such adaptations are referred to as Personalized PageRank (PPR) variants, which can be expressed in matrix form as follows [43]:

$$\mathbf{r} = \alpha \mathbf{T} \mathbf{r} + (1 - \alpha) \mathbf{d} \quad (5)$$

For PPR, $\frac{1}{N} \mathbf{1}_N$ from equation 4 becomes $\mathbf{d}$ in equation 5, representing a personalized bias vector. This adjustment means that probability mass is no longer uniformly distributed across all nodes, but instead is distributed to reflect an encoded bias. Similarly, edge weights can be incorporated by adjusting the transition matrix $\mathbf{T}$ influencing how probability mass is redistributed among nodes at each iteration. Some notable PPR variants include:

- Topic-sensitive PageRank [52], which creates separate topic-biased vectors, rather than relying on a single uniform distribution. For each topic vector, additional probability mass is directed toward a subset of sources relevant to that theme, identified using a word taxonomy. Queries (words) are matched to one or more topic-specific vectors, and final scores are generated by linearly combining these topic vectors with weights based on query–topic similarity. An additional α parameter controls the mixture between the topic-biased and uniform vectors.
- CiteRank [53], which introduces a temporal bias vector, weighting nodes according to an exponential time-decay function and giving more probability to recent publications than older ones. This adjustment reflects how researchers tend to discover and cite newer work more readily, capturing the dynamics of attention toward recent papers.
- TrustRank [43], which modifies PageRank to combat web spam by assigning initial probability only to a small set of manually chosen ‘trusted’ seed nodes. Over iterations, trust (probability mass) propagates outward along links, dampened by distance from the seed nodes.
- Weighted PageRank [54], which distributes a node's probability mass to its outlinks proportional to the destination nodes' importance, rather than splitting it

evenly. Link importance is based on both the inlink popularity and outlink connectivity of the destination node, so nodes with many inlinks or outlinks receive a larger share of probability mass.

2.3.3 Information diffusion models

Information diffusion models describe how ideas, behaviors, or influence spread throughout large-scale networks. They typically assume that information originates at one or more seed nodes, and once initialized, they spread this information to their direct neighbors. Once a node has received this piece of information, it in turn spreads the information to its direct neighbors.

Diffusion models vary in how nodes transmit information to neighbors [55]. In the independent cascade model [56] active nodes (nodes with a piece of information) have a single chance at each time step to activate inactive neighbors, with success determined by a fixed probability. Newly activated nodes then attempt to activate their own neighbors, and the process continues until no further activations occur. Such models are often used to identify subsets of nodes that maximize information or influence spread in social networks [57].

Epidemic models meanwhile treat information spread like disease transmission, with the two common variants being the susceptible-infective-susceptible (SIS) and the susceptible-infective-recovered (SIR) models [58]. These describe the possible states nodes can take in a network: infected nodes hold the information and susceptible nodes can acquire information. In SIS, infected nodes can become susceptible again, while in SIR, infected nodes can recover and 'become immune' to information. These and related models have been applied to rumor propagation [59], social media cascades such as retweets on X (formerly twitter) [33] [55], and to identifying emerging fields in scientific disciplines [60]. Notably, these models sometimes identify influential nodes different from those highlighted by centrality measures [61].

2.4 Measures of influence and attention outside of network science

2.4.1 Traditional bibliometric measures

Outside of network science, there are several commonly used measures of a researcher or publication's impact, particularly in academic contexts:

1. The h-index, developed in 2005 by Hirsch [62], is a widely used measure of author productivity and impact, based on both publication output and citation counts. It is calculated by ranking all an author's publications in descending order of citation count, and identifying the maximum value h such that the researcher has published h papers, each of which has been cited at least h times. The index mitigates the influence of single, highly cited works as well as uncited publications, offering a more balanced assessment of scholarly impact [63]. However, the h-index needs to be contextualized, as citation counts can vary depending on the subject area and time since publication [63].
2. The g-index, developed in 2006 by Egghe [64], was proposed as an enhancement of the h-index to provide a more comprehensive measure of the overall citation performance of a set of publications. When the publications are ranked in descending order by citation count, the g-index is defined as the largest number g such that the top g articles have collectively received at least g^2 citations. The g-index gives more weight to highly cited publications, but is less widely used than the h-index [65].
3. Clarivate's journal impact factor (JIF) is a journal ranking system, where high-impact journals are considered more prestigious and important in their field of study [66]. JIF reflects the average frequency with which articles from a journal are cited within a given year. It is calculated as the number of citations in the current year to items published in the preceding two years, divided by the total number of citable items published by the journal during that same period.

The above measures are rarely used in conjunction with network science approaches to determining influence and impact. For example, Zhou et al. [67] proposed a h-index-based method for predicting missing links in a citation network. Network science

approaches are most often used to create more nuanced methods for evaluating impact and influence that consider relational dynamics [68], or to predict measures like h-indexes [69].

2.4.2 Altmetrics

Altmetrics (alternative metrics) have emerged since around 2010 to capture dimensions of impact and engagement beyond traditional academic citations [70]. Altmetrics track mentions and engagement across diverse platforms such as news outlets, policy documents, patents, blogs, social media platforms like X, Reddit, Wikipedia, YouTube and reference managers like Mendeley [71]. *Altmetric.com* is one of the main databases of altmetric data, as well as others like *Impactstory* and *PlumX Metrics*. Going forward, references to altmetrics or altmetric data will refer to data sourced from *Altmetric.com*.

Altmetric.com gives publications an attention score, reflecting how much attention a research publication receives online, particularly across social media [72]. The more widely discussed or shared a publication is, the higher its attention score. This score is calculated as a weighted count, where different sources of attention are assigned different values (ranging from 0.25 to 8). News outlets and blogs receive the highest weight, while other channels such as social media posts contribute smaller weights. The system also applies modifiers, for example, mentions from a major national newspaper adds more to the score than a local or highly specialized outlet [73].

Scores and attention sources are updated daily, highlighting immediate and real-time attention to a publication and its visibility outside academia. This contrasts to conventional metrics, which accumulate slowly over longer time periods and primarily reflect scholarly uptake. Another aspect of the *Altmetric.com* attention score is that it is not a measure of quality, rather of how much attention the publication receives. This also differs from traditional metrics which are taken as proxies of a publication's quality.

Citation network studies largely focus on homogeneous structures within a network, e.g., solely on academic publications, or on social media posts. Even when heterogeneity is introduced, it is usually restricted - e.g., academic ecosystems with

author–journal–paper links - without incorporating wider domain sources such as policy documents, news media, or other online content. There is room to broaden citation networks to include a wider range of altmetric sources to better understand research dissemination [74].

3 Methodology

This study introduces a variant of PPR tailored to measuring attention hazards, termed Multiplicative PPR (MPPR). Unlike most citation network studies, which focus on mapping the influence of, or identifying high-impact papers, MPPR is designed to help assess risk relevance and the amplification of dual-use information. To date, methods such as PageRank have not been applied in dual-use research and information hazards contexts.

In academic contexts, citation-based studies often emphasize the accuracy, quality, or context of citations. For attention hazards however, the relevance of a citation or mention lies in its function as a conduit for attention to a given idea. While altmetrics have increasingly complemented traditional citation measures, they remain underused in network-based analyses. MPPR addresses this by incorporating *Altmetric.com*-like attention scores to model heterogeneous amplification pathways beyond academia, highlighting the role of policy, media, and other non-academic amplifiers in bringing attention to information hazards (see section 3.2.3 and **Appendix A**). This study also integrates diverse altmetric sources into the network alongside academic publications. Although social media posts are excluded here, their inclusion could provide further nuance in future analyses of heterogeneous citation networks.

Although the network analysis approaches and measures outlined in the Literature Review are often used on large-scale networks, they can equally be applied to ego-centric networks, as these are subgraphs of a larger global citation network [75]. Traditional centrality measures such as degree, betweenness, or eigenvector centrality treat all nodes and edges as equally relevant, overlooking factors like time, influence pathways, or thematic alignment with core topics. In an ego-centric citation network limited to three concentric layers (including core nodes), purely structural ranking risks overemphasizing hubs or well-connected nodes that may not be relevant to the diffusion of IH information. While RWBC could address some structural bias of an ego-centric network, it is designed only for undirected graphs and is therefore unsuitable for directed citation networks.

By contrast, MPPR's flexible bias mechanism can encode a variety of domain-specific priors, beyond proximity to core papers or node recency to things like topical relevance to information hazards. This ensures PageRank scores remain sensitive to structural connectivity while reflecting the dynamics most relevant to the study at hand. The damping term α mitigates the problem of boundary nodes acting as probability mass sinks when they lack outgoing links. Some boundary nodes may have outgoing links however if cited elsewhere in the network.

Epidemic-style diffusion models such as SIS or SIR are well suited to capturing stochastic retransmission on platforms like X, where information may spread globally within minutes or hours. By contrast, citations create fixed retrospective links: once a source cites another, the edge is permanent and does not propagate further like a contagion. Citation dynamics also unfold on much slower timescales of months or years.

Epidemiological parameters such as transmission probability or recovery have no clear equivalence in citation networks. In the localized ego-centric network constructed here all nodes are already directly or indirectly linked to the core research papers, leaving no susceptible or unexposed nodes to be 'infected'. This fundamentally contrasts with epidemic models, which assume a global network structure containing both infected and uninfected node populations.

MPPR meanwhile provides a computationally efficient and probabilistically interpretable framework that respects the static, directional properties of citation data, building on methods widely applied in citation network studies to capture influence and visibility. Its stationary distribution represents the long-run probability of residing at a given node, capturing how attention gradually accumulates and information is amplified across the network.

3.1 Core information hazard-related papers

MPPR builds on prior work with PPR, adapting it to measure attention hazards by tracking how information from five core biotechnology papers diffuses outward through their first-order (one hop) and second-order (two hop) sources. In this setup, MPPR

scores act as a proxy for attention: the more probability mass a node accumulates, the greater its role in amplifying information from core papers of dual-use concern and in drawing attention to them. These landmark papers illustrate the dual-use dilemma in virology and biotechnology, where scientific advances sparked substantial public and academic debate over their potential for misuse:

- In 2001, Jackson et al. [14] engineered mousepox by inserting a gene for the immune system hormone interleukin-4, intending to sterilize rodents by triggering antibodies against egg proteins. Instead, the modification made the virus more lethal and able to evade immunity, even in vaccinated mice [76]. Being related to the human smallpox virus, the study raised fears that a similar tweak could undermine the human smallpox vaccine [77]. The organization funding the study issued a press release urging stronger safeguards against bioweapons emerging from well-intentioned research [76].
- Cello, Paul, and Wimmer [15] achieved chemical synthesis of a live poliovirus in 2002, proving that viruses could be created without natural templates. Given its publication less than a year after 9/11 and the 2001 anthrax attacks in the United States, public concerns were raised that other deadly viruses could also be created [15, 78].
- Both Herfst et al. [16] and Imai et al. [17] independently showed in 2012 that avian influenza H5N1 could be engineered for airborne transmission between mammals (ferrets), igniting intense debate over the publication of gain-of-function research [79], where organisms gain new or enhanced functions through engineered genetic modifications [80]. The United States National Science Advisory Board for Biosecurity reviewed both studies to determine their pandemic potential for humans [81]; while ultimately published, key methodological details needed to replicate the experiments were redacted [79].
- In 2018, Noyce et al. [18] synthesized horsepox virus entirely from DNA fragments sourced from the internet. It was able to infect cells and reproduce, demonstrating the feasibility of assembling poxviruses de novo and raising

questions about barriers to reconstructing extinct or eradicated pathogens like smallpox [82, 83].

3.2 Network structure

The above papers serve as core nodes in an ego-centric network, a portion of a global network organized around specific entities known as egos [84]. Nodes directly or indirectly linked to these egos are called alters [85].

3.2.1 Layers

The localized network of these cores papers stops at two hops from core nodes. While a node may be within two hops of one core paper and further from another (e.g., if it cites a second-order source linked to a different core paper), the boundary ensures that all nodes are at most two steps from at least one core. This two-hop cutoff aligns with previous findings that network influence typically decays sharply beyond three hops [86, 87], suggesting that such local networks capture most of the meaningful dynamics of spread and amplification from a given node. More intuitively, two hops mark the limit at which an idea can still be easily traced back to a core paper. For example, an individual encountering a second-order source could follow its bibliography to a first-order source citing the core paper, and from there reach the core itself. Beyond two hops, however, the connection to the core paper becomes increasingly diluted.

3.2.2 Edge direction

In constructing the citation network, edge direction is inverted such that an edge $(v_i, v_j) \in E$ denotes ‘document v_i is cited by document v_j ’. This reverses the standard citation network set up where v_i cites v_j , as it better aligns the network with the study’s focus on information flow and amplification. Under conventional edge orientation, applying MPPR to the ego-centric network would cause the core papers to dominate probability mass share, as all citation paths terminate at them. Since PageRank concentrates probability mass in high in-degree nodes, the cores would absorb most of the stationary distribution - an unsurprising outcome that merely restates their centrality in an ego-centric network, offering little analytical value.

By inverting edge direction, the network models the outward diffusion of information from core papers to their alters. Probability mass flows from core nodes to their citing nodes and beyond, so MPPR highlights those that receive and propagate this information rather than concentrating solely on the cores. The stationary distribution thus captures the reach of information into surrounding layers, identifying intermediaries that amplify visibility of this information. This adjustment is particularly important for analyzing attention hazards, as it treats citations and mentions as forward-propagating channels through which IH content gains traction. Note however that inlinks to core papers are possible if they cite other sources in the network.

3.2.3 Nodes

The network contains several node types, each assigned an attention score based on *Altmetric.com*'s framework which measures how much attention different source types attract [19]. For simplicity, only base attention scores are adapted for this study. Score modifiers like those used by *Altmetric.com* to distinguish between high and low-reach entities within a source type (e.g. national vs local news outlets) are excluded [88]. A full list of node types, their attention scores, and score justifications is provided in **Appendix A**. Attention scores for citing sources determine edge weights in MPPR (see section 3.3.4).

3.3 MPPR

MPPR incorporates modifications to both the bias vector $\mathbf{d}$ and transition probability matrix $\mathbf{T}$ in equation 5 (section 2.3.2). In matrix form, it is expressed as:

$$\mathbf{r} = \alpha \mathbf{P}^T \mathbf{r} + (1 - \alpha) \mathbf{b} \quad (6)$$

Where:

- $\mathbf{r}$ is an $n \times 1$ vector of MPPR scores (the stationary distribution).
- $\mathbf{P}^T$ is the transpose of the weighted transition matrix, so that probability mass flows from cited $\rightarrow$ citing (see section 3.2.2).
- $\mathbf{b}$ is the bias vector (a biased probability distribution over all nodes).
- α is the damping factor, set to 0.85.

3.3.1 Layer score

To measure the structural proximity of nodes to the core papers, each node i is assigned a continuous layer score $L_i \in [0,1]$. The score is derived from a sequence of PPR computations, once per core node $c \in \mathcal{C}$, run on an unweighted version of the citation graph with damping factor $\alpha = 0.85$.

In each computation run, the initial probability distribution assigns full weight to the core node c and zero to all others. The resulting stationary distribution quantifies how strongly every node in the graph is connected to c . Let $PPR_i^{(c)}$ denote the stationary probability of node i in a computation run centered on core node c . To prevent a core node from artificially inflating its own proximity score ($PPR_c^{(c)}$), its score is manually set to zero after convergence. The resulting vectors from each computation run are then averaged, producing a single proximity score that reflects each node's overall accessibility from the entire core set:

$$AvgPPR_i = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} PPR_i^{(c)} \quad (7)$$

Finally, proximity scores for nodes are min-max normalized over all nodes N so that the final layer score L_i for a given node lies in $[0,1]$ and is comparable in scale to the time-decay factor (see section 3.3.2):

$$L_i = \frac{AvgPPR_i - \min_{(j \in N)} AvgPPR_j}{\max_{(j \in N)} AvgPPR_j - \min_{(j \in N)} AvgPPR_j} \quad (8)$$

This construction of the layer score has two advantages. First, it captures multi-path, accessibility from the entire core set, rather than relying solely on shortest-hop distances. Second, averaging across cores prevents bias toward any single core while excluding self-influence. The resulting L_i provides a smooth and interpretable measure of structural proximity to the core papers, scaled consistently with other model features.

3.3.2 Time decay

In addition to structural proximity, temporal dynamics are incorporated using an exponential time-decay function that down-weights older publications relative to newer ones, the same form used in CiteRank [53]. For each node i , its age is calculated as the difference between the current year and its publication year: $age_i = y_{\text{now}} - y_i$. The corresponding decay factor is given by:

$$D_i = e^{-\lambda \cdot age_i}, \quad \text{where } age_i = y_{\text{now}} - y_i, \text{ and } \lambda > 0 \quad (9)$$

where $\lambda > 0$ is a tunable decay parameter that controls how quickly influence decreases with age. Recent sources receive values close to 1, while older sources are assigned progressively smaller weights, reflecting the decline in attention over time. The exponential form provides a smooth, proportional decrease, avoiding arbitrary cutoffs while ensuring recency strongly shapes the probability distribution.

To parameterize the exponential time-decay function, the decay constant λ is derived from empirically observed half-lives of citation accrual and URL link decay. Reported values include: Walker et al. [53], ~2.6 years for preprints and 1 year for journals; Habibzadeh [89], 2.2 – 5.3 years (mean of 3.75) for website references in academics journals; Goh and Ng [90], ~5 years for links to information science journals; and for general website URL availability, ~2 years [91].

From these figures, an average half-life is calculated to be 2.87 years. Formally, if the expected number of new citations at age t is proportional to $e^{-\lambda t}$, then the half-life $t_{1/2}$ is defined as the time when this rate falls to half its initial value, i.e.

$$e^{-\lambda t_{1/2}} = 0.5$$

Solving for λ gives:

$$\lambda = \frac{\ln 2}{t_{1/2}} \quad (10)$$

From the average half-life calculated above, we obtain $\lambda \approx 0.24 \text{ yr}^{-1}$, meaning that the rate of new citations declines by 50 % approximately every 2.9 years.

3.3.3 Bias vector

The final MPPR bias value for node i is:

$$b_i = \frac{L_i D_i}{\sum_j L_j D_j} \quad (11)$$

Each node's bias is determined by the product of its layer score L_i and time-decay factor D_i . The numerator gives higher weight to more recent nodes that are closer to core papers, while the denominator normalizes the distribution so that all probabilities sum to 1.

This differs from PPR variants such as Topic-Sensitive PageRank (section 2.3.2). In that approach, multiple topic-specific bias vectors are defined (e.g., sports, technology, health), and a separate PPR run is executed for each. The resulting outputs are then linearly combined to form a weighted average PPR value across contexts, but at the cost of maintaining and mixing multiple vectors.

By contrast, MPPR uses a single bias vector that directly integrates both structural proximity (layer score) and temporal recency (time decay). Rather than running multiple PPR computations, these two dimensions are combined multiplicatively into one probability distribution. This yields a simpler model - similar in approach to CiteRank - yet one that amplifies nodes that are **both structurally and temporally relevant** in ways that other PPR variants do not.

3.3.4 Modification of T with edge weights

The other novel feature of MPPR is the use of edge weights to bias the transition probability matrix. Edge weights represent how much attention citations attract according to node type. As discussed in section 3.2.3, the network contains several node types, each type assigned a base attention score reflecting relative visibility and reach, similar to *Altmetric.com*'s framework (see **Appendix A**).

Attention scores for citing nodes determine the weights of their outgoing edges: citations from more visible or authoritative sources receive higher weights (reflecting more attention) than those from lower-visibility sources.

Mathematically, edge weights are incorporated into the transition matrix. For node i with outgoing neighbors $j \in N(i)$, the transition probability of moving along edge $i \rightarrow j$ is given by

$$P(i \rightarrow j) = \frac{\omega_{ij}}{\sum_{k \in N(i)} \omega_{ik}} \quad (12)$$

where w_{ij} is the assigned edge weight. The denominator ensures probabilities sum to 1 over all neighbors.

3.4 Identification of information hazard-related sources

To assess a source's level of engagement with IH content and dual-use concerns, a keyword and key phrase-based lookup scoring method is applied. This method provides a simple, transparent, and interpretable way to identify nodes mentioning topics of interest such as dual-use biotechnology, deliberate misuse, or governance of high-consequence research. While this section focuses on the scoring methodology, details regarding data processing and cleaning steps underpinning this method are outlined in section 4.2.

Each node receives a score reflecting the strength of its connection to IH biotechnology discourse, based on the number of matched IH concepts in collected node metadata (e.g., titles, summaries, assigned topics). Concepts are grouped into tiers with corresponding scores:

1. **Critical terms** (+8 each): e.g., “gain of function,” “information hazard,” “bioterrorism,” “weapons of mass destruction,” “military application”.
2. **High-relevance terms** (+5 each): e.g., “biosecurity threat,” “biodefense,” “information security,” “synthetic biology”,
3. **Medium-relevance terms** (+2 each): e.g., “risk assessment,” “national security,” “chemical warfare,” “proliferation,” “governance”, and
4. **Broad terms of potential interest** (+1 each): e.g., generic terms such as “risk” or “regulation” that require further context to be considered IH concepts.

A special case applies if a node contains exactly one critical term and no others. It receives an additional +8 to emphasize singular but highly salient flags (e.g., a title “Gain-of-function research policy”). Refer to **Appendix B** for the full list of terms.

The final output is a continuous score, with higher values indicating greater relevance to biotechnology IH concerns. Contributing terms are retained to maintain an audit trail. Due to the method’s simplicity, it may generate false positives or negatives depending on IH term coverage and metadata quality. To reduce errors, a threshold score of ≥ 7 is used to classify nodes as IH, requiring a match of at least one high-relevance and one medium-relevance term. More nuanced methods such as NLP models (e.g., word2vec) could help parse longer text snippets, produce a richer set of IH terms, and pick up the relevance of these terms in the wider context of the source.

4 Experiments

4.1 Data collection

Data to create the citation graph are collected from three different databases'

Application Programming Interfaces (APIs), accessed through a custom Python script:

1. OpenAlex API - academic articles, books, book chapters, datasets, dissertations, editorials, errata, grants, letters, paratexts, peer-reviews, reference entries, reports and reviews [92-94].
2. Altmetric.com's Details API - blog posts, guidelines, news websites, patents, policy documents, video media and Wikipedia articles [95, 96].
3. Overton's REST API - clinical guidance, periodicals, policy documents, academic articles, transcripts, and working papers [97].

4.1.1 OpenAlex data

The pipeline retrieves first-order citations, as well as second-order citations from OpenAlex and Overton first-order sources. OpenAlex exposes explicit citation links, enabling reference chains across multiple orders of citation. For each core paper's Digital Object Identifier (DOI) the script resolves its OpenAlex ID, queries all citing sources, and stores their metadata in structured comma-separated values (CSV) files. Collected fields include identifiers (OpenAlex ID, DOI), bibliographic information (title, publication date, type), citation API URL, primary topic (domain, subfield), associated topics, keywords, concepts, and MeSH descriptors.

Second-order extraction proceeds by querying all sources that cite each first-order OpenAlex source, saving the same metadata plus the cited first-order IDs. Results are grouped into five CSV files, one per core paper. Additionally, DOIs from Overton sources are resolved to OpenAlex IDs, and their citing sources are retrieved. These second-order results are stored in one combined CSV file, with metadata and cited first-order Overton DOIs.

4.1.2 Altmetric.com data

The pipeline retrieves first-order citations, and where possible, second-order citations from OpenAlex and Overton first-order sources with DOIs. Second-order data could not be collected from *Altmetric.com* first-order sources, as the Details API does not expose links between *Altmetric.com* source IDs or URLs and other sources that cite them.

Queries to the Details API use DOIs from core papers and first-order sources. For each DOI, the script resolves an *Altmetric.com* ID and retrieves detailed “posts” metadata, filtered to relevant online sources (blogs, video, news, policy, guideline, patent, Wikipedia). Metadata for each mention - source type, title, URL, posting date, and summary - are cleaned and stored in standardized CSV files. For second-order sources, the corresponding first-order sources they cite are also recorded. A combined CSV file is produced for second-order sources derived from Overton first-order sources.

4.1.3 Overton data

The pipeline retrieves Overton first-order sources, as well as second-order citations from Overton, OpenAlex, and *Altmetric.com* first-order sources. Queries to the Overton API use DOIs from core papers, Overton IDs from first-order sources, DOIs extracted from OpenAlex first-order sources, and DOIs extracted from *Altmetric.com* mentions.

For each core paper DOI, the script queries Overton for all citing sources, saving metadata in one CSV file per core paper. Metadata fields include: Overton ID, title, translated title (for non-English sources), document type, publication date, document URL, source tags, top topics, languages, document themes, and any associated DOIs.

Second-order extraction is performed in three ways:

1. From Overton IDs - first-order Overton IDs are queried to retrieve all citing Overton sources, with results stored in CSVs grouped by first-order sources per core paper.
2. From OpenAlex DOIs - DOIs extracted from OpenAlex first-order sources are queried in Overton, with citing sources saved to five CSVs.

3. From *Altmetric.com* DOIs - policy and patent mentions from *Altmetric.com* are title-matched in OpenAlex to resolve DOIs, which are then queried against Overton; results are saved to five CSVs.

4.2 Data processing

4.2.1 Data Cleaning

The data cleaning pipeline standardizes and consolidates raw collection data into consistent, analysis-ready CSVs for node and edge list creation. Across all sources, columns are mapped to a common schema, dates are standardized, and DOIs are normalized. Where possible, DOIs replace database-specific to enable cross-database linking.

OpenAlex. Sources are deduplicated by DOI and type. For first-order sources, if a DOI appears with multiple types (e.g., preprint and journal article), only the earliest record is retained; if duplicates exist for the same DOI-type pair, the first entry is kept. For second-order sources, records are grouped by second-order DOI, first-order DOI, and type. If duplicates share the same second- and first-order DOIs but differ by type, the earliest record is retained; if both DOIs and type match, the first entry is kept. This ensures each second-order source maps uniquely to its cited first-order source.

Altmetric.com. Language detection (via *langdetect* [98]) removes non-English entries unless a translated title or English summary is available for keyword/phrase matching (section 3.4). For deduplication, rows without unique identifiers are dropped, and duplicates by identifier are resolved by retaining the first entry. For second-order data, duplicate pairs of second- and first-order IDs are reduced to a single entry, ensuring unique second–first order links.

Overton. Non-English titles are replaced by translated versions where available; otherwise, entries lacking sufficient metadata (e.g., top topics) for keyword/phrase matching are removed. Overton IDs are treated as unique, and duplicates are resolved by retaining the first record. The same rule applies to DOIs when present. For second-order sources derived from OpenAlex or *Altmetric.com*, duplicate pairs of second-order

ID (Overton ID or DOI) and cited first-order DOI are resolved by retaining the first entry, again ensuring unique links.

Finally, the cleaned outputs are compiled into grouped files by dataset and citation layer (e.g. *compiled_openalex_first_order.csv*, *compiled_altmetric_second_order.csv*).

4.2.2 Creating edges and nodes lists

4.2.2.1 Edge list

Edge list construction begins by harmonizing identifiers and building a global ID → node type map (e.g., *policy_document*, *academic_article*). All IDs are normalized, and node type labels extracted from compiled files are stored for later use in edge labeling and weighting. Metadata for the core papers are fetched from OpenAlex, saved to a CSV file, and added to the global map under the label *core_paper*.

With this scaffolding in place, first-order → core paper edges are created, labeling each pair as source (citing) and target (cited) node. Node type labels are attached to both source and target nodes. Second-order → first-order links are then mapped using processed IDs from the cleaning step. Each batch filters targets against a first-order whitelist (per database) to prevent stray or missing nodes from multi-dataset stitching appearing in the final graph. Node type labels are again assigned via the global map.

Finally, all edges are concatenated into a single data frame. Type labels are standardized to a controlled vocabulary (e.g., *blogs/blog post* → *blog_post*). Edges are weighted according to the source node's attention score (**Appendix A**). Duplicate source-target pairs are resolved by keeping one pair, and the cleaned edge list is saved with the following fields: *source_id*, *target_id*, *source_type*, *target_type*, *edge_type*, *weighting*.

4.2.2.2 Node list

To identify IH nodes, source-specific text fields are concatenated prior to concept extraction: *title*, *document_theme*, and *top_topics* for Overton; *title*, *primary topic*, *topics*, *keywords*, *concepts*, and *MeSH* for OpenAlex; and *title* and *summary* for *Altmetric.com*. These fields are pre-processed to ensure consistent matching across sources.

Concept detection proceeds in two passes. First, multi-word expressions (MWEs) from a curated lexicon (e.g., *dual-use*, *gain of function*) are matched, including common orthographic variants (e.g., *dual use* → *dual-use*), with all hits mapped to a canonical form. Second, atomic keywords (e.g., *biodefense*, *biorisk*) are detected using boundary-aware regex patterns. To prevent double counting, atomic terms subsumed by MWEs are excluded. A secondary canonicalization step then standardizes spelling and morphological variants (e.g., *weaponisation* → *weaponization*, *infohazards* → *information hazard*), again resolving duplicates of already-matched canonical concepts. The result is a deduplicated set of canonicalized IH concepts per record, which underpins the scoring procedure in Section 3.4.

The three database source-specific data frames are concatenated into one data frame, and duplicate rows resolved where required. Any missing core papers and metadata are added. Temporal and structural cleanup is then applied in alignment with the edge list: publication dates are normalized, unparseable rows dropped, and nodes older than the earliest core paper removed. Edges connected to pruned nodes are deleted, except where the counterpart node still connects elsewhere in the graph. Fully disconnected nodes are dropped, and edges are validated to ensure both endpoints exist in the node list. Node types are then extracted by mapping source and target node IDs to edge list labels.

Final outputs include a nodes file with 92,586 unique entries (*id*, *title*, *publication_date*, *infohazard_concepts*, *score*, *node_type*) and an edges file with 107,129 unique edges.

4.3 Graph construction

Graphs are built using the Graph-tool python module [99]. Graph-tool was selected because its core data structures and algorithms are implemented in C++, giving performance comparable to pure C/C++ libraries. In contrast, Python-native packages such as NetworkX or igraph run too slowly on graphs of the size constructed here [100]. A limitation is that Graph-tool runs only on Mac or Linux systems, however its flexibility in modifying graph structures and algorithm parameters made it preferable to alternative packages.

Nodes and edges files are used to construct the main directed graph for analysis. Each node is assigned properties for: ID, title, publication year (parsed from `publication_date`, with missing values imputed to the dataset mean year), node type, IH score, and a boolean “infohazard related” flag (True if score is ≥ 7 . or if the node is a core paper). Edges are built in the reversed direction (cited $\rightarrow$ citing; see section 3.2.2) and assigned weights based on the type of the cited node under this orientation.

Because Graph-tool requires separate structures for different analyses, additional graph variants are created to help with evaluation methods (section 4.4): one with the original citation orientation citing $\rightarrow$ cited, and another undirected, unweighted version (see Section 4.4).

4.4 Evaluation methods

4.4.1 Main comparative analysis

To evaluate the performance and robustness of MPPR, a series of comparative analyses are conducted against a baseline: a directed graph matching MPPR edge orientation, with unweighted edges and a uniform bias vector (**UNW+UBV**). This baseline is equivalent to standard unweighted PageRank and serves as a neutral benchmark for assessing how alternative weighting and bias strategies affect ranking outcomes. The following variants were tested:

1. **MPPR** - Weighted graph, MPPR bias vector.
2. **UNW+MBV** – Unweighted graph, MPPR bias vector.
3. **Rev+MBV** – Weighted graph, reversed edges (citing $\rightarrow$ cited), MPPR bias vector.
4. **W+UBV** – Weighted graph, uniform bias vector.
5. **W+RBV** – Weighted graph, random bias vector.
6. **UUW+UBV** – Undirected and unweighted graph, uniform bias vector.
7. **W+LBV** – Weighted graph, bias vector restricted to layer scores.
8. **W+DBV** – Weighted graph, bias vector restricted to time decay.
9. **W+MWBV_{mw value}** – Weighted graph, mixed weight bias vectors (layer-time decay combination, where *mw* controls layer-biased vector weight).

10. **W+TBV** – Weighted graph, bias vector restricted to node type (node type weighting usually only applies to edge weights).

Variant 9 is the equivalent approach of TopicSensitive PageRank, where bias vectors (layer score and time decay) are precomputed and combined linearly:

$$MWBV = mw \times \text{layer vector values} + (1 - mw) \times \text{time decay vector values}$$

All variants are evaluated relative to the baseline UNW+UBV using a unified set of metrics:

- i. Spearman rank-order correlation coefficient (ρ) between each variant and the baseline [101]. This nonparametric measure, which does not assume any underlying distribution, assesses the strength and direction of monotonic relationships between two ranked variables, in this case ranked node PageRank scores. Values range from -1 (perfect negative monotonic association) to $+1$ (perfect positive monotonic association), with 0 indicating no association.
- ii. Top-k overlap ($k = 100, 500, 1\%, 5\%$). This measures the percentage of nodes in common with the baseline among the top-ranked sets.
- iii. Percent of IH nodes in the top-k subsets ($k = 100, 500, 1\%, 5\%$).
- iv. Average percentile rank (APR) for IH nodes, reported in absolute terms and as deltas (change) relative to the baseline. Each node's percentile rank is calculated as [102]:

$$\text{Percentile rank} = 100 \times \left(\frac{\text{rank} - 1}{N} \right) \quad (13)$$

where *rank* is the node's position when sorted by PageRank score (1 = highest, N = lowest), and N is the total number of nodes. This rescales ranks to $[0,100)$, so the top-ranked node has a percentile of 0 , and the lowest-ranked percentile approaches 100 (but never reaches it). For example, a node with a percentile rank of 25 will be ranked higher than 75% of all nodes. The APR of a group (e.g., IH nodes) is the arithmetic mean of their individual percentile ranks, summarizing their position within overall distribution.

- v. Statistical separation tests between IH and non-IH nodes using the Mann–Whitney U test with effect size reported as Cliff’s delta (δ).

The Mann–Whitney U test is a nonparametric method for comparing two independent samples [103, 104]. Here, the two groups are the percentile ranks of IH nodes and non-IH nodes. The null hypothesis (H_0) is that both groups are drawn from the same distribution; the alternative (H_1) is that their distributions differ, such that IH nodes systematically occupy higher or lower percentile ranks. The test is appropriate for PageRank percentiles, which are often skewed and heavy-tailed.

The U statistic is derived by first ranking all observations from lowest to highest, and ties are handled by assigning them the average of the ranks they would otherwise occupy. The rank sums are then calculated separately for each group. From these, U is computed as the number of times an observation from one group ranks higher than an observation from the other, with tied values split evenly between groups. Here, U reflects how often IH nodes rank above or below non-IH nodes in percentile terms, adjusted to account for ties.

Given the large sample size ($N = 92,586$), p-values are not emphasized as even negligible differences become statistically significant, and p-values tend to shrink, as the sample size grows [105, 106]. Instead, practical significance is assessed via effect size. The commonly used $r = |z|/\sqrt{N}$ effect size formulation for the U statistic [103] is avoided here, as it is sensitive to both the total size of the sample and unequal sizes between groups [107]. In this study IH nodes make up only 4.34% of the total data.

Instead, Cliff’s delta (δ) provides a robust, nonparametric measure of stochastic dominance between groups. It is defined as the probability that a randomly chosen node from one group has a higher percentile rank than a node from the other group, minus the reverse probability. This is calculated by comparing all possible pairs of observations between groups. The result is divided by the product of the sample sizes of the two groups, expressed as [108]:

$$\delta = \frac{\#(x_{i1} > x_{j2}) - \#(x_{i1} < x_{j2})}{n_1 n_2} \quad (14)$$

where # denotes the count of instances, x_{i1} is an observation from group 1, x_{j2} is an observation from group 2, and n_1, n_2 are the respective group sizes. The values range from -1 to +1, with the extremes indicating no overlap between the two distributions, and 0 indicating complete overlap. Effect sizes are generally interpreted as negligible ($|\delta| < 0.15$), small ($0.15 < |\delta| < 0.33$), medium ($0.33 < |\delta| < 0.47$) and large ($|\delta| > 0.47$) [108]. However, classification needs to be considered carefully as what a small or large effect size means in practice is dependent on the context. A small or negligible effect size can still lead to meaningful and measurable changes in the data distributions between the two groups under comparison.

Computing δ directly requires all $n_1 \times n_2$ pairwise comparisons, which is computationally intensive at scale [108]. A more efficient approach uses the Mann-Whitney U statistic. Vargha and Delaney's A (VDA) is defined as the probability that a randomly chosen node from group 1 ranks higher than a node from group 2, while splitting ties evenly between the groups [109]. It can be obtained by scaling the U statistic [110]:

$$VDA = \frac{U}{n_1 n_2} \quad (15)$$

δ is then a linear transformation of VDA [109, 111, 112]:

$$\begin{aligned} \delta &= 2 \times VDA - 1 \\ &= \frac{2U}{n_1 n_2} - 1 \end{aligned} \quad (16)$$

To complement evaluation metrics (i-v) above, each variant is visualized through plots of IH and non-IH node PPR score distributions and percentile rank probability density

distributions. These visualizations reveal distributional features such as spread, skew, and overlap that may not be captured by the summary metrics alone.

4.4.2 Experiment 1

Experiment 1 evaluates how different edge-weighting schemes affect ranking stability, baseline overlap, and IH node enrichment. Experiment 1a tests edge weighting across varying values of α (the damping parameter, see section 2.3.2 and equation 3), while 1b assesses the sensitivity of each variant to changes in α relative to a fixed baseline.

4.4.2.1 1a: edge weight impact on model performance

Experiment 1a evaluates how alternative edge-weighting schemes affect ranking outcomes across different values of $\alpha \in \{0.99, 0.85, 0.50\}$. Metrics assessed include:

- i. Spearman rank-order correlation with the baseline,
- ii. Top-K overlap ($K = 100, 500, 1\%, 5\%$),
- iii. Percent of IH nodes in top-K subsets, and
- iv. APR of IH nodes, in both absolute terms and relative deltas to the baseline.

The standard α value of 0.85 is used as a reference point, with higher α values placing more emphasis on edge-following, and lower values placing more weight on the bias distribution. The baseline variant used is UNW+MBV, and the time-decay parameter is fixed at $\lambda = 0.24$.

MPPR is then run on the baseline and with the following edge-weight formulations at each α :

- **BW**: Base weights reflecting attention scores in **Appendix A** for citing nodes;
- **BW_TD**: $\text{BW} \times \text{time-decay values}$,
- **BW_TD_L**: $\text{BW_TD} \times \text{layer scores}$,
- **BW_TD_L_T**: $\text{BW_TD_L} \times \text{additional weights for } \textit{cited} \text{ nodes types (Appendix A)}$.
- For each of the above, three further variants apply an exponent $\beta \in \{3.0, 6.0, 9.0\}$.

These tests evaluate whether incorporating richer edge weight information alters ranking stability, overlap with the baseline, or IH node prominence. Exponential scaling

magnifies weight differences, highlighting whether stronger emphasis on high-scoring nodes shifts results.

4.4.2.2 1b: sensitivity analysis for α

Experiment 1b evaluates how sensitive each edge-weighting scheme is to changes in α . Using $\alpha = 0.85$ as a reference baseline, each scheme from Experiment 1a is compared against its own results at alternative α values, evaluated using metrics i-iv in 1a.

Whereas Experiment 1a highlights performance differences between edge-weighting schemes under fixed conditions, Experiment 1b captures within-scheme variation as the balance between structural influence (edges) and bias distribution is adjusted by α .

4.4.3 Experiment 2

Experiment 2 seeks to understand IH node representation in top-K subsets and by node type under different modeling choices. This experiment is divided into two parts.

4.4.3.1 Experiment 2a: enrichment of IH nodes

Experiment 2a examines how IH node representation among top-K subsets changes under different weighted variants with the same edge direction. A baseline score is established using the MPPR model, and results are compared against the following variants:

- W+UBV
- W+RBV
- W+LBV
- W+DBV
- W+MWBV_25
- W+MWBV_50
- W+MWBV_75
- W+TBV

For each variant, enrichment analysis is conducted relative to MPPR by measuring the change (Δ) in the proportion of IH nodes within the top-K subsets ($K = 100, 500, 1\%, 5\%$).

This provides a diagnostic of how sensitive IH node enrichment is to alternative modeling assumptions about the bias vector.

4.4.3.2 Experiment 2b: PageRank probability mass share gains and losses by node type and IH status

Experiment 2b investigates how probability mass is redistributed across node types when moving from a baseline model (UNW+UBV) to MPPR. For each configuration, PageRank scores are aggregated by node type and stratified by IH status (IH vs. non-IH).

Mass share for each group is defined as the proportion of total PageRank probability mass assigned to that category. Differences are calculated as both absolute changes and relative percentage changes between the baseline and MPPR. This highlights which node categories, and which IH statuses, gain or lose influence under MPPR.

4.4.4 Experiment 3

To investigate factors associated with amplification or dampening of IH sources under MPPR, Experiment 3 compares PageRank results between a baseline model (UNW+UBV) and MPPR. For each node, the change in score between models is computed as ΔPR . Three explanatory variables are considered:

- i. mean base edge weight (average of all outgoing edge weights for a node),
- ii. layer score, and
- iii. time-decay factor.

Correlation analysis is then performed between ΔPR and each variable restricted to IH nodes. Scatter plots illustrate the relationships, while both Pearson's correlation coefficient (r) and Spearman's rank correlation coefficient (ρ) are reported. Pearson's r captures linear associations on continuous data [113], while Spearman's ρ provides a robustness check for monotonic but potentially non-linear relationships.

5 Results and discussion

5.1 Main comparative analysis

Table 1: Main comparative analysis – PPR variants vs baseline (UNW+UBV)

Variant	Spear- Man ρ	Overlap top_100 (%)	Overlap top_500 (%)	Overlap top 1pct (%)	Overlap top 5pct (%)	Top_100 IH_pct (%)	Top_500 IH_pct (%)	Top 1pct_IH (%)	Top 5pct_IH (%)	Δ top 100 (%)	Average IH percentile rank (AIHPR)	Δ AIHPR (%)	Mann- Whitney U	Mann- Whitney p-value	Effect size (δ)
Baseline	1	100	100	100	100	20	16	13.73	9.85	0	41.88	0	1.48E+08	1.35E-74	-0.17
MPPR	0.76	20	26.8	28.86	45.97	24	23.8	23.24	13.31	4	44.12	2.24	1.57E+08	5.97E-40	-0.12
W+UBV	0.94	76	81.2	82.16	81.83	17	13.6	12.86	8.34	-3	43.52	1.64	1.54E+08	3.91E-48	-0.14
UNW+MBV	0.76	19	26.6	28.65	46.23	25	23.6	23.46	13.48	5	43.98	2.11	1.56E+08	9.78E-42	-0.13
W+RBV	0.05	31	39	29.62	16.42	12	11.2	10.27	5.68	-8	48.54	6.66	1.73E+08	1.02E-03	-0.03
W+LBV	0.87	29	24.8	31.89	35.86	41	43.8	33.08	19.42	21	35.45	-6.43	1.24E+08	4.14E-235	-0.3
W+DBV	0.06	29	25	22.49	11.77	9	5.4	4.97	2.31	-11	64.82	22.95	2.34E+08	6.65E-244	0.31
W+MWBV_ 0.25	0.19	30	32.4	32.86	38.71	37	33.8	31.57	16.79	17	49.89	8.01	1.78E+08	8.07E-01	0
W+MWBV_ 0.5	0.35	31	26.2	32.11	34.85	39	38.2	32.97	19.08	19	46.52	4.64	1.66E+08	4.83E-15	-0.07
W+MWBV_ 0.75	0.56	29	25	31.78	35.93	41	42.6	33.08	19.36	21	42.67	0.79	1.51E+08	3.82E-61	-0.15
W+TBV	0.64	4	3.2	2.81	4.3	14	6.4	7.03	4.56	-6	45.37	3.49	1.61E+08	2.11E-25	-0.1
Rev+MBV	-0.01	5	5.2	8.32	12.14	14	22.2	13.3	7.65	-6	50	8.19	1.78E+08	7.12E-01	0
UUW+UBV	-0.12	0	0.4	0.32	11.8	11	15	15.35	15.66	-9	48.36	6.48	1.72E+08	2.28E-04	-0.03

Figure 1: Distributions of PPR scores per variant (baseline = UNW+UBV)

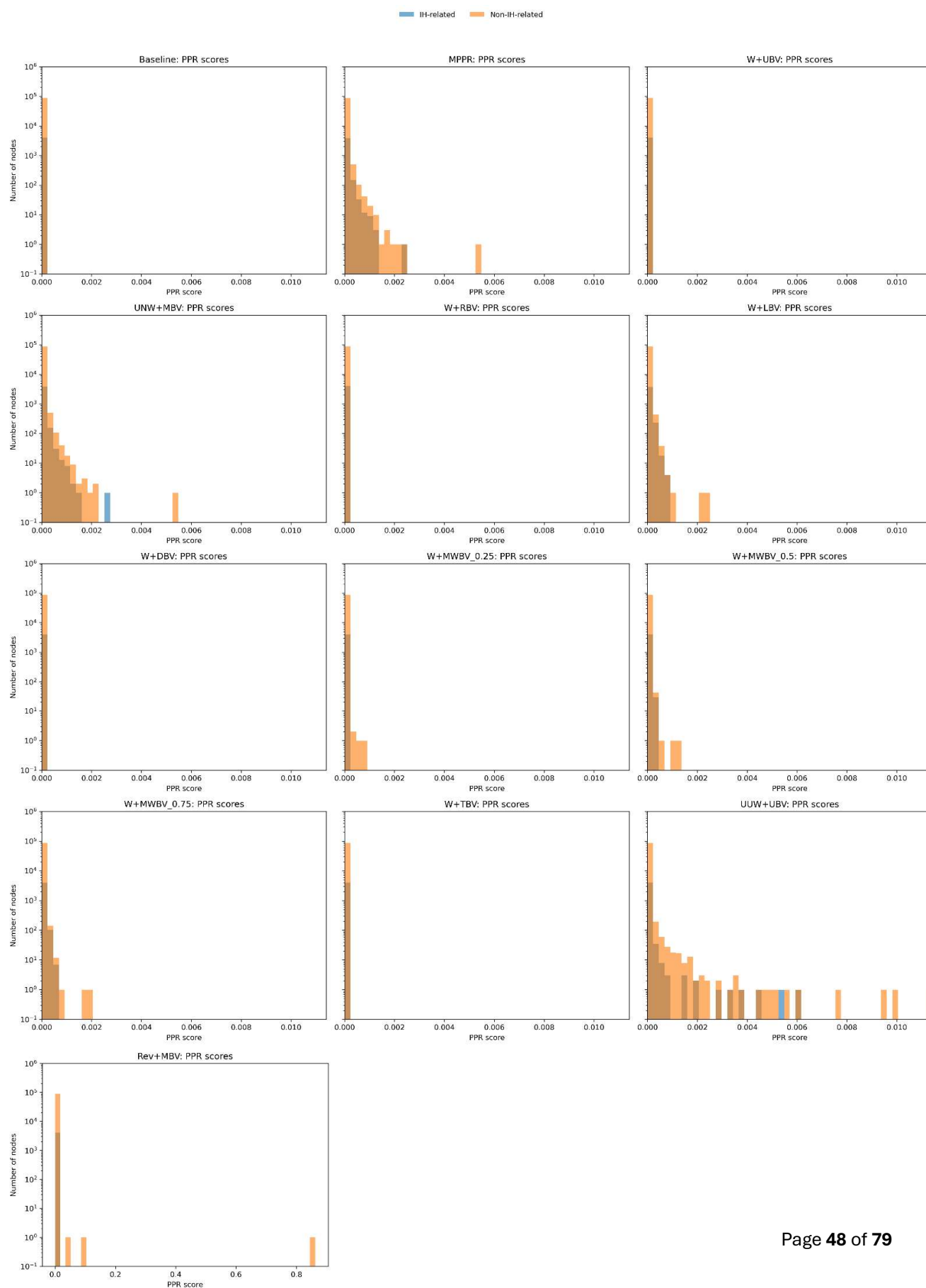
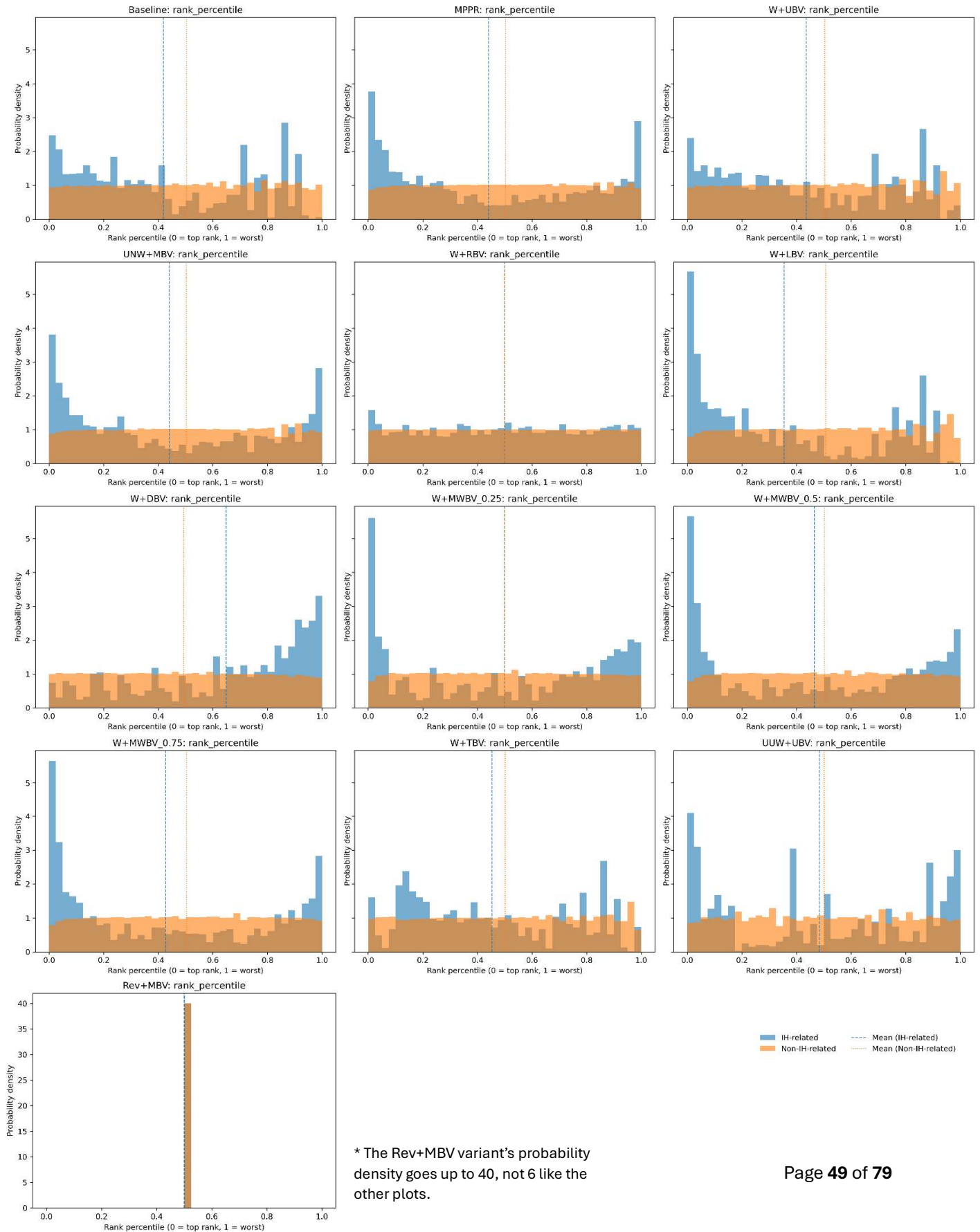


Figure 2: Probability density distributions per variant of average percentile rank for IH and non-IH nodes (baseline = UNW+UBV)



5.1.1 Trends in top-K subsets

As shown in Table 1, MPPR diverges from the baseline (UNW+UBV) in overall rankings ($\rho = 0.76$), with even sharper divergence in the top-K subsets (overlap of only 20%, 26.8%, 28.86%, and 45.97% for K=100, 500, 1%, 5% respectively). For IH nodes, MPPR moderately increases their presence in top-K subsets and improves their APR by 2.24%.

Relative to other variants, MPPR performs as a middle ground approach. Weighted variants with bias vectors tied to single features or mixed weighting (W+LBV, W+DBV, W+MWBV) generally show higher overlap with the baseline in top-K subsets (except at the top 5%). W+TBV is an outlier, with very low overlap with the baseline. In terms of IH node representation, W+LBV and W+MWBV consistently over-represent IH nodes, with W+MWBV converging toward W+LBV as weight shifts toward the layer score. By contrast, W+DBV and W+TBV under-represent IH nodes, with W+DBV performing worst overall (AIHPR = 64.82). All variants except W+DBV have higher IH node representation at K= 5% than the whole network IH node representation (4.34%).

5.1.2 Effect sizes

MPPR's balanced behavior is further reflected in the effect size δ , which quantifies separation between IH and non-IH rank distributions and contextualizes the Mann-Whitney U statistic.³ Because lower percentiles indicate better ranks, negative δ values mean IH nodes are favored, while positive values mean they are disadvantaged.

In the baseline, $\delta = -0.17$, showing IH are modestly prioritized over non-IH nodes. W+LBV produces a stronger negative effect size ($\delta = -0.30$), close to the threshold of a medium effect size, consistent with the fact it has the lowest AIHPR (35.5) of any variant and overrepresents IH nodes in top subsets. By contrast, W+DBV shows the opposite tendency ($\delta = 0.31$), indicating IH nodes are systematically pushed down, consistent with it having the highest AIHPR value.

With MPPR's $\delta = -0.12$. This supports the view of MPPR as a middle ground approach, neither overinflating nor suppressing IH node rankings relative to the baseline.

³ As discussed in section 4.4.1, p-values for the Mann-Whitney U statistic are not emphasized as even negligible differences become statistically significant, and p-values tend to shrink, as the sample size grows. P-values have been included in Table 1 to demonstrate this effect.

Importantly, Table 1 shows that even variants with negligible size effects are clearly having a measurable impact on node rankings and IH node representation at top-K subsets, despite distributions of IH and non-IH nodes remaining largely overlapping.

5.1.3 Impact of variants on data distributions

5.1.3.1 PageRank score distributions

Figure 1 shows that most directed graph variants, aside from MPPR, yield highly compressed PageRank score distributions. Nearly all nodes cluster around very small values (~ 0.000 – 0.001), with only a few standing out in the tail. Among single-feature bias variants, W+LBV and W+MWBV (with higher weightings toward layer scores) produce somewhat broader distributions. MPPR produces the widest spread among weighted directed variants, while UNW+MBV shows a broader spread than most weighted models. While Rev+MBV exhibits the largest range, but this is driven by a few extreme outliers rather than a genuine spread. U UW+UBV produces the broadest distribution overall; removing both directionality and weighting flattens structural asymmetries and allows probability mass to diffuse more evenly across the network. This results in less concentration and more nodes accumulating relatively higher stationary probabilities.

MPPR's broader spread among weighted variants arises from its non-uniform bias structure. In the baseline, bias is uniform, so all 92,586 nodes receive approximately the same tiny baseline probability mass, limiting variation and preventing outliers. Variants with distributions resembling the baseline similarly fail to generate differentiated probability mass shares across nodes.

MPPR-style bias vectors are effective at redistributing probability mass among nodes because their bias is hardcoded to two conditions: layer score and time decay. This bias is amplified through propagation across the network, so favored nodes (newer nodes with higher layer scores) accumulate probability mass disproportionately, while disfavored nodes lose mass they would retain under the baseline. The result is sharper contrasts between nodes and a wider spread of scores. Mixed-weight variants, by contrast, compress the distribution. Their post-hoc averaging of completed PageRank vectors bounds each node's score between values in the two components, pulling high

scores downward and low scores upward. In this setup, W+LBV forms the upper bound and W+DBV the lower bound.

5.1.3.2 Percentile rank density distributions

Figure 2 shows how variants affect the density distribution of IH and non-IH nodes across percentile ranks. A striking feature of Figure 2 is the relative stability of the non-IH node distributions. Aside from in Rev+MBV, non-IH nodes spread evenly across the percentile rank spectrum, forming a relatively flat baseline. This reflects their numerical dominance (88,552 nodes), which populates the full ranking range and smooths the distribution. As a result, differences between variant distributions arise less from reordering non-IH nodes, and more from shifting the much smaller IH node subset against this background.

IH nodes meanwhile deviate strongly from uniformity. In the baseline, their density is slightly skewed toward higher ranks, creating modest separation from non-IH nodes. Under MPPR, IH nodes form a U-shaped distribution with a slight right skew, meaning concentrations of nodes appear at both extremes with fewer in the middle. Variants such as UNW+MBV, W+LBV, and W+MWBV show similar patterns, although with varying degrees of right skew. W+LBV and W+DBV have the most pronounced skews, shifting the IH mean furthest from the non-IH mean and producing the strongest effect sizes.

A point to note is that if IH nodes made up a much larger share of the network, additional adjustments would be needed to avoid distortion in interpreting rankings. A higher base rate would mean that raw top-K percentages might reflect prevalence rather than genuine prioritization by a variant. The flat baseline formed by the dominant non-IH population in Figure 2 illustrates this risk - if IH prevalence increased, raw top-K shares could overstate prioritization effects.

5.1.3.3 Conventional citation direction

The Rev+MBV variant illustrates the effect of running PageRank on the ego-centric network with the conventional citation direction (citing → cited). As Figure 1 illustrates, nearly all nodes collapse to near-zero values with only a few outliers, while Figure 2 shows high densities of both IH and non-IH node percentile ranks with values of ~0.5. This occurs because probability mass flows inward to high in-degree nodes - in this

case, the five core papers. Since all citation paths terminate at the cores, the stationary distribution assigns negligible weight to other nodes, flattening their relative rankings. The result is that PageRank merely reconfirms the cores' importance and provides little insight into attention diffusion. By contrast, inverting edge direction redistributes probability mass outward from the cores, producing meaningful variation in scores and highlighting intermediary nodes that propagate information. This underscores the critical role of edge orientation in modeling information flow.

5.1.4 Unweighted vs weighted variants

In the main comparative analysis, the baseline and MPPR can be compared with W+UBV and UNW+MBV respectively, to assess the impact of edge weights on variant performance. W+UBV shows the highest ρ value and top-K overlap with the baseline (Table 1), with minimal differences in IH representation. UNW+MBV results are nearly identical to MPPR, and Figures 1 and 2 confirm that weighted and unweighted variants of UBV and MPPR yield very similar PageRank and percentile rank density distributions. This suggests that edge weights have little influence on overall rankings, regardless of variant. Their effects are examined in greater detail in Experiment 1.

5.2 Experiment 1

The results of Experiment 1a and 1b are presented in **Appendix C** (Tables 2 and 3). Experiment 1a evaluates the effect of edge weight modifications relative to a baseline (UNW+MBV). Table 2 shows that across all α values, base weights (BWs) used in MPPR have little impact on rankings. Node overlap in top-K subsets changes only slightly, IH node presence remains stable, and the AIHPR shifts by just 0.126% on average. This aligns with findings in Section 5.1.4, where minimal differences are observed between weighted and unweighted versions of UBV and MPPR.

Even when weights are inflated by incorporating time decay, layer scores, target node types, or exponential scaling, the effects remain surprisingly limited. Overall ρ values vary by no more than 0.01 across all tested variants. Top-K overlaps generally decline as more features or higher exponents are applied, but changes in IH node representation

or AIHPR are negligible. The largest shift is -6% in IH node representation among the top 100 nodes; all other IH node measures deviate by no more than 2% from the baseline.

Experiment 1b (Table 3) compares each variant at $\alpha \in \{0.99, 0.85, 0.50\}$ against its own baseline at $\alpha = 0.85$. ρ values remain 1.0 across all cases, while top-K overlaps decrease modestly as more feature information, stronger exponents, or larger deviations in α are applied. The lowest overlap is 85% for the top 100 nodes. Effects on IH node representation are similarly minor - the largest shift is a -7% in IH nodes among the top 100, and the largest change in AIHPR is -0.28.

The limited influence of edge weighting across Experiments 1a and 1b likely stems from the network's ego-centric structure. Because it is bounded around five core nodes and their first- and second-order sources, the topology is highly constrained; all paths originate from the cores, and most terminate in two layers. With likely very few inlinks to the cores from other nodes, and mostly short paths available, probability mass has fewer routes to redistribute. Edge weight modifications may adjust flow along these fixed paths, but they cannot introduce new ones or override the dominance of the core-source hierarchical structure. Consequently, even exaggerated edge-weighting schemes produce only marginal differences in overlap, IH node presence, and AIHPR.

5.3 Experiment 2

Table 4: Experiment 2a - Enrichment of information hazard-related nodes (baseline = MPPR)

Variant	IH % (Top 100)	IH % (Top 500)	IH % (Top 1%)	IH % (Top 5%)	Δ IH % (Top 100)	Δ IH % (Top 500)	Δ IH % (Top 1%)	Δ IH % (Top 5%)
MPPR	24	23.8	23.33	13.3	0	0	0	0
W+UBV	17	13.6	12.86	8.34	-7	-10.2	-10.47	-4.96
W+RBV	9	10.8	9.19	5.64	-15	-13	-14.14	-7.66
W+LBV	41	43.8	33.08	19.42	17	20	9.75	6.12
W+DBV	9	5.4	4.97	2.31	-15	-18.4	-18.36	-10.99
W+MWB V_0.25	37	33.8	31.53	16.8	13	10	8.2	3.5
W+MWB V_0.5	39	38.2	32.94	19.07	15	14.4	9.61	5.77
W+MWB V_0.75	41	42.6	33.15	19.35	17	18.8	9.82	6.05
W+TBV	14	6.4	7.03	4.56	-10	-17.4	-16.3	-8.74

Experiment 2a (Table 4) compares weighted variants with the same edge orientation against MPPR. When IH node percentages across top-K subsets are averaged for all variants excluding MPPR (26.25, 24.28, 20.65, and 11.97 for K= 100, 500, 1%, 5%), the results are close to MPPR’s own values (24, 23.8, 23.33, 13.3). This convergence occurs because the variants tend to diverge in opposite directions: W+LBV and W+MWBV consistently elevate IH nodes into higher-ranked subsets, while W+DBV, W+RBV, and W+TBV suppress them. Aggregated together, these upward and downward shifts offset one another, yielding averages that approximate MPPR’s more neutral distribution. This supports the earlier finding (Section 5.1.1) that MPPR acts as a “middle ground” variant, balancing the more extreme biases of alternative bias vector strategies.

Table 5: Experiment 2b - PageRank probability mass share gains and losses by node type and IH status (MPPR vs baseline = UNW+UBV)

Node type	IH status	Node count	Probability mass share baseline	Probability mass share MPPR	Δ Probability mass share	Probability mass relative change (%)
book	TRUE	12	1.58E-04	2.70E-03	2.54E-03	1613.55
paratext	TRUE	2	2.23E-05	2.68E-04	2.46E-04	1104.77
news_website	TRUE	223	2.47E-03	1.97E-02	1.72E-02	697.31
policy_document	TRUE	179	2.19E-03	1.66E-02	1.45E-02	661.43
book_chapter	TRUE	476	5.25E-03	1.80E-02	1.28E-02	243.52
wikipedia	TRUE	30	3.24E-04	1.10E-03	7.74E-04	239.06
video	TRUE	4	4.33E-05	1.34E-04	9.10E-05	209.1
report	TRUE	9	1.06E-04	3.26E-04	2.20E-04	207.15
editorial	TRUE	46	5.01E-04	1.38E-03	8.80E-04	175.62
academic_article	TRUE	1887	2.08E-02	5.67E-02	3.60E-02	173.2
review	TRUE	909	9.87E-03	2.46E-02	1.48E-02	149.54
policy_document	FALSE	1632	1.84E-02	3.84E-02	2.00E-02	108.92
blog_post	FALSE	1265	1.40E-02	2.79E-02	1.39E-02	99.05
blog_post	TRUE	60	6.64E-04	1.24E-03	5.78E-04	86.94
book	FALSE	238	2.66E-03	4.96E-03	2.30E-03	86.46
editorial	FALSE	445	4.88E-03	8.62E-03	3.74E-03	76.57
other	TRUE	58	6.43E-04	1.05E-03	4.07E-04	63.32
book_chapter	FALSE	4810	5.21E-02	8.13E-02	2.92E-02	56.16
video	FALSE	54	5.90E-04	9.04E-04	3.14E-04	53.25
dissertation	TRUE	10	1.08E-04	1.60E-04	5.10E-05	47.45
report	FALSE	48	5.24E-04	7.51E-04	2.27E-04	43.38
other	FALSE	556	5.97E-03	8.02E-03	2.05E-03	34.26
wikipedia	FALSE	366	3.97E-03	5.06E-03	1.09E-03	27.42
core_paper	TRUE	5	5.27E-05	6.61E-05	1.30E-05	25.28
letter	TRUE	43	4.76E-04	5.27E-04	5.10E-05	10.64
paratext	FALSE	113	1.22E-03	1.27E-03	5.80E-05	4.74
patent	FALSE	2177	2.36E-02	2.42E-02	6.03E-04	2.56
letter	FALSE	880	9.54E-03	9.59E-03	4.90E-05	0.51
news_website	FALSE	7885	8.43E-02	7.94E-02	-4.95E-03	-5.87
erratum	FALSE	22	2.41E-04	2.10E-04	-3.10E-05	-12.92
preprint	FALSE	6083	6.54E-02	5.55E-02	-9.90E-03	-15.15
preprint	TRUE	78	8.35E-04	6.66E-04	-1.69E-04	-20.2
clinical_guidance	FALSE	182	1.97E-03	1.57E-03	-4.04E-04	-20.47
review	FALSE	13112	1.41E-01	1.09E-01	-3.16E-02	-22.43
academic_	FALSE	47748	5.15E-01	3.93E-01	-1.22E-01	-23.76

Node type	IH status	Node count	Probability mass share baseline	Probability mass share MPPR	Δ Probability mass share	Probability mass relative change (%)
article						
periodical	FALSE	4	4.61E-05	3.16E-05	-1.50E-05	-31.48
dissertation	FALSE	554	5.94E-03	3.48E-03	-2.46E-03	-41.46
peer-review	FALSE	214	2.29E-03	1.05E-03	-1.25E-03	-54.35
guideline	FALSE	19	2.07E-04	8.98E-05	-1.17E-04	-56.63
working_paper	FALSE	32	3.43E-04	1.48E-04	-1.96E-04	-56.94
reference-entry	TRUE	1	1.06E-05	3.41E-06	-7.00E-06	-67.84
reference-entry	FALSE	26	2.78E-04	7.69E-05	-2.01E-04	-72.33
dataset	FALSE	81	8.66E-04	1.15E-04	-7.50E-04	-86.66
grant	FALSE	1	1.13E-05	1.35E-06	-1.00E-05	-88.02
retraction	FALSE	1	1.06E-05	2.84E-07	-1.00E-05	-97.32
legal documents	FALSE	3	3.20E-05	4.96E-07	-3.10E-05	-98.45
clinical_guidance	TRUE	1	1.08E-05	1.26E-07	-1.10E-05	-98.83
press release	FALSE	1	1.05E-05	1.11E-07	-1.00E-05	-98.95
patent	TRUE	1	1.06E-05	6.33E-08	-1.00E-05	-99.4

Table 5 shows how probability mass is redistributed under MPPR relative to the UNW+UBV baseline in Experiment 2b, disaggregated by node type and IH status. The vast majority of IH nodes gain probability mass, with decreases limited to four node types and very small counts (the largest being 78 IH preprints, while the others involve just one node each).

Most gains occur in source types with higher attention scores (**Appendix A**). For both IH and non-IH groups, blogs (score 5), Wikipedia (5), videos (5), policy documents (3), books and book chapters (2), and reports (2) all increase in mass. Among news websites (score 8), IH nodes show some of the largest gains (+697.31%), while non-IH nodes show a slight decline (-5.87%). Clinical guidance is the only high-scoring source type (3) to decline for both IH and non-IH nodes. Overall, high-attention node types that gained in at least one IH subgroup account for ~66.3% of the total positive mass shift between MPPR and the baseline, despite representing only ~54% of gaining node types.

The broad increase in IH probability mass under MPPR reflects the alignment between IH node characteristics and the features emphasized in MPPR's bias vector. MPPR favors recent sources that directly cite core papers, while IH nodes are identified independently through keyword and phrase counts in titles, summaries, and topic tags.

This means that many IH sources must fall into node types that share these features. As shown in Experiment 1 and the main comparative analysis, edge weights based on attention scores (**Appendix A**) have little influence on rankings. Thus, the observed gains for IH nodes are driven primarily by MPPR's bias vector design rather than by attention-based edge weighting.

An important caveat in interpreting these results is that the large relative gains recorded for IH node types partly reflect their small subgroup sizes. Under the baseline, probability mass is spread almost uniformly across the 92,586 nodes, so IH subgroups begin with very low baseline shares. When MPPR redistributes attention according to temporal and layer-based biases, many IH categories align with these features and gain mass. Yet because their starting values are so small, modest absolute increases translate into very large relative changes (e.g., +1613.55% for IH books, +661.43% for IH policy documents). By contrast, non-IH subgroups in the same categories, often numbering in the thousands, start from larger baseline shares, so comparable or larger absolute gains appear proportionally smaller. Thus, while MPPR structurally favors IH categories aligned with its biases, the magnitude of relative changes should be interpreted with caution, as they are amplified by small denominators.

5.4 Experiment 3

Table 6: Experiment 3 – Correlation analysis between features and change in PageRank scores between MPPR and baseline (UNW+UBV) for IH nodes

Feature	Pearson's r	r p-value	Spearman's ρ	ρ p-value	Number of IH nodes
Base weight (mean)	0.21	1.25E-39	0.38	1.41E-139	4034
Time decay	0.36	1.35E-120	0.55	3.494e-318	
Layer score	0.63	0.00E+00	0.8	0.00E+00	

Figure 3: Experiment 3 – Scatter plots between features and change in PageRank scores between MPPR and baseline (UNW+UBV) for IH nodes

Scatter plots: features vs Δ PageRank (MPPR – baseline) on IH-related nodes

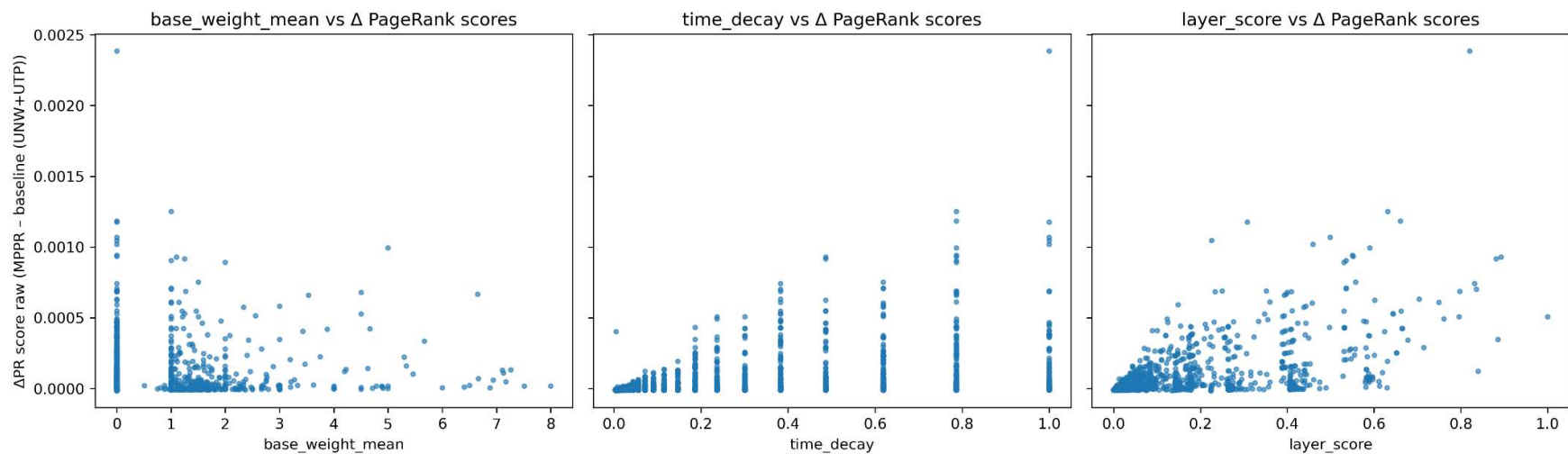


Table 6 reports correlations between select features and changes in PageRank scores for IH nodes ($n = 4034$) when comparing MPPR with a UNW+UBV baseline. At this scale, even weak associations yield very small p-values, making significance tests uninformative. The analysis therefore focuses on the size and direction of the correlation coefficients (r and ρ) as indicators of substantive relationships between node features and PageRank score changes.

The results show that base weights (derived from attention scores in **Appendix A**) are only weakly correlated with changes in IH node PageRank scores (Pearson's $r = 0.21$, Spearman's $\rho = 0.38$), suggesting edge weights play only a modest role in redistribution. Time decay shows a stronger correlation ($r = 0.36$, $\rho = 0.55$), indicating that recency is more influential in whether IH nodes gain or lose prominence under MPPR. The strongest relationship is with layer score ($r = 0.63$, $\rho = 0.80$), showing that IH nodes positioned closer to the core papers gain disproportionately under MPPR.

The scatter plots in Figure 3 reinforce these findings. For base weights, most nodes cluster at low values with minimal change, and no clear relationship emerges between higher weights and PageRank shifts. The time decay plot shows a clear upward trend: newer IH nodes (higher decay scores) are more likely to gain probability mass, consistent with recency as a key driver of attention. The layer score plot shows the strongest linear trend, with nodes closer to the cores capturing larger gains. Overall, these plots confirm that MPPR's redistribution for IH nodes is driven primarily by structural position (layer) and temporal recency (time decay), while base edge weights contribute comparatively little.

6 Conclusion

This dissertation has shown that MPPR provides a robust framework for quantifying biotechnology attention hazards, and offers a formal mechanism for studying how information hazards are amplified within heterogeneous citation networks. By integrating structural position and temporal recency into its bias design, MPPR redistributes probability mass in line with expected amplification dynamics: public-facing and policy-relevant sources like blogs, websites, videos, and policy documents, gain most visibility and therefore attention, while specialist academic sources are comparatively suppressed. This suggests that the prominence of IH nodes under MPPR reflects their structural and temporal alignment with the model's biases, rather than their substantive content.

The analysis also highlights important limitations. First, MPPR does not explain why certain sources are more influential than others; for example, the role of an author's public profile or organizational affiliations in amplifying attention is not captured. Nor does it account for external factors that may shape the risk profile of a source, such as whether the information requires a high threshold of technical knowledge to be used in malicious dual-use applications. These dimensions remain important avenues for future research.

Second, IH nodes constitute a small minority of nodes in this study, but if their prevalence were to rise substantially, raw top-K percentages could reflect base rates rather than true model prioritization. Without safeguards, variant comparisons would lose interpretive value. Future modeling could address this by adapting evaluation methods - for example, comparing observed versus expected IH node counts under IH label-shuffle baselines to test whether prioritization exceeds chance.

Finally, this dissertation underscores that MPPR is best understood as a flexible platform rather than a finished solution. Its framework can readily incorporate advances such as NLP methods to refine IH node identification, or more sophisticated edge-weighting schemes tested on diverse network topologies. These extensions could strengthen the model's ability to separate genuine amplification effects from artefacts of structure or prevalence. The results produced by the model, particularly in

Experiment 2b, equip policymakers and researchers with a more effective tool for anticipating and mitigating biotechnology attention hazards.

7 Bibliography

- [1] N. Bostrom, "Information hazards: A typology of potential harms from knowledge," *Review of Contemporary Philosophy*, no. 10, pp. 44-79, 2011.
- [2] D. Rotolo, D. Hicks, and B. R. Martin, "What is an emerging technology?," *Research Policy*, vol. 44, no. 10, pp. 1827-1843, 2015, doi: <https://doi.org/10.1016/j.respol.2015.06.006>.
- [3] T. Ord, *The Precipice: Existential Risk and the Future of Humanity*. London: Bloomsbury Publishing Plc, 2020.
- [4] N. Bostrom, "Existential Risk Prevention as Global Priority," *Global Policy*, vol. 4, no. 1, pp. 15-31, 2013, doi: <https://doi.org/10.1111/1758-5899.12002>.
- [5] I. Carrozza, N. Marsh, and G. M. Reichberg, "Dual-Use AI Technology in China, the US and the EU: Strategic Implications for the Balance of Power," Peace Research Institute Oslo, Oslo, 2022. [Online]. Available: <https://www.prio.org/publications/13150>
- [6] R. M. Atlas and M. Dando, "The Dual-Use Dilemma for the Life Sciences: Perspectives, Conundrums, and Global Solutions," *Biosecurity and Bioterrorism: Biodefense Strategy, Practice, and Science*, vol. 4, no. 3, pp. 276-286, 2006, doi: 10.1089/bsp.2006.4.276.
- [7] G. Lewis, P. Millett, A. Sandberg, A. Snyder-Beattie, and G. Gronvall, "Information Hazards in Biotechnology," *Risk Analysis*, vol. 39, no. 5, pp. 975-981, 2019, doi: <https://doi.org/10.1111/risa.13235>.
- [8] M. Ienca and E. Vayena, "Dual Use in the 21st Century: Emerging Risks and Global Governance," *Swiss Medical Weekly* vol. 148, no. 4748, pp. 1-8, 2018, doi: <https://doi.org/10.4414/smw.2018.14688>.
- [9] A. Hunter, "Synthetic Biology and National Security: Risks and Opportunities (Part 1 of 2)," ed: Centre for Strategic and International Studies, 2020.
- [10] A. Radford et al., "Better Language Models and Their Implications," OpenAI, 14 February 2019. [Online]. Available: <https://openai.com/blog/better-language-models/>
- [11] F. Urbina, F. Lentzos, C. Invernizzi, and S. Ekins, "Dual use of artificial-intelligence-powered drug discovery," *Nature Machine Intelligence*, vol. 4, no. 3, pp. 189-191, 2022/03/01 2022, doi: 10.1038/s42256-022-00465-9.
- [12] M. J. Selgelid, "Governance of dual-use research: an ethical dilemma," (in en), *Bulletin of the World Health Organization*, vol. 87, no. 9, pp. 720-723, 2009, doi: 10.2471/BLT.08.051383.
- [13] A. Dubov, "The concept of governance in dual-use research," *Medicine, Health Care and Philosophy*, vol. 17, no. 3, pp. 447-457, 01/08/2014, doi: 10.1007/s11019-013-9542-9.
- [14] R. J. Jackson, A. J. Ramsay, C. D. Christensen, S. Beaton, D. F. Hall, and I. A. Ramshaw, "Expression of Mouse Interleukin-4 by a Recombinant Ectromelia Virus Suppresses Cytolytic Lymphocyte Responses and Overcomes Genetic Resistance to Mousepox," *Journal of Virology*, vol. 75, no. 3, pp. 1205-1210, 2001, doi: 10.1128/jvi.75.3.1205-1210.2001.
- [15] J. Cello, A. V. Paul, and E. Wimmer, "Chemical Synthesis of Poliovirus cDNA: Generation of Infectious Virus in the Absence of Natural Template," *Science*, vol. 297, no. 5583, pp. 1016-1018, 2002, doi: 10.1126/science.1072266.
- [16] S. Herfst et al., "Airborne Transmission of Influenza A/H5N1 Virus Between Ferrets," *Science*, vol. 336, no. 6088, pp. 1534-1541, 2012, doi: 10.1126/science.1213362.
- [17] M. Imai et al., "Experimental adaptation of an influenza H5 HA confers respiratory droplet transmission to a reassortant H5 HA/H1N1 virus in ferrets," *Nature*, vol. 486, no. 7403, pp. 420-428, Jun. 2012, doi: 10.1038/nature10831.
- [18] R. S. Noyce, S. Lederman, and D. H. Evans, "Construction of an infectious horsepox virus vaccine from chemically synthesized DNA fragments," *PLOS ONE*, vol. 13, no. 1, p. e0188453, 2018, doi: 10.1371/journal.pone.0188453.

- [19] "How is the Altmetric Attention Score calculated?" Altmetric.com. Accessed: 28 August 2025 [Online]. Available: <https://help.altmetric.com/support/solutions/articles/6000233311-how-is-the-altmetric-attention-score-calculated->.
- [20] T. Hagendorff, "Forbidden knowledge in machine learning reflections on the limits of research and publication," *AI & SOCIETY*, vol. 36, no. 3, pp. 767-781, Sep. 2021, doi: 10.1007/s00146-020-01045-4.
- [21] P. Torres, "Agential risks and information hazards: An unavoidable but dangerous topic?," *Futures*, vol. 95, pp. 86-97, Jan. 2018, doi: <https://doi.org/10.1016/j.futures.2017.10.004>.
- [22] K. M. Esvelt, "Inoculating science against potential pandemics and information hazards," *PLOS Pathogens*, vol. 14, no. 10, p. e1007286, 2018, doi: 10.1371/journal.ppat.1007286.
- [23] B. D. Trump, M.-V. Florin, E. Perkins, and I. Linkov, Eds. *Emerging Threats of Synthetic Biology and Biotechnology: Addressing Security and Resilience Issues* (NATO Science for Peace and Security Series C: Environmental Security). Lausanne, Switzerland Springer Dordrecht, 2021.
- [24] A. Casadevall, T. S. Dermody, M. J. Imperiale, R. M. Sandri-Goldin, and T. Shenk, "On the Need for a National Board To Assess Dual Use Research of Concern," *Journal of Virology*, vol. 88, no. 12, pp. 6535-6537, 2014, doi: 10.1128/jvi.00875-14.
- [25] A. Ovadya and J. Whittlestone, "Reducing malicious use of synthetic media research: Considerations and potential release practices for machine learning," *arXiv preprint arXiv:1907.11274*, 2019.
- [26] T. Shevlane and A. Dafoe, "The Offense-Defense Balance of Scientific Knowledge: Does Publishing AI Research Reduce Misuse?," presented at the Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society, New York, NY, USA, 2020. [Online]. Available: <https://doi.org/10.1145/3375627.3375815>.
- [27] A. Brenneis, "Assessing dual use risks in AI research: necessity, challenges and mitigation strategies," *Research Ethics*, vol. 21, no. 2, pp. 302-330, 2025, doi: 10.1177/17470161241267782.
- [28] E. H. Soice, R. Rocha, K. Cordova, M. Specter, and K. M. Esvelt, "Can large language models democratize access to dual-use biotechnology?," *arXiv preprint arXiv:2306.03809*, 2023.
- [29] D. Hendrycks, M. Mazeika, and T. Woodside, "An overview of catastrophic AI risks," *arXiv preprint arXiv:2306.12001*, 2023.
- [30] F. Eiras *et al.*, "Risks and opportunities of open-source generative AI," *arXiv preprint arXiv:2405.08597*, 2024.
- [31] Y. Zeng *et al.*, "AI Risk Categorization Decoded (AIR 2024): From Government Regulations to Corporate Policies," *SuperIntelligence - Robotics - Safety & Alignment*, vol. 1, no. 1, 10 Sep. 2024, doi: 10.70777/si.v1i1.10603.
- [32] D. Knoke and S. Yang, "Introduction to Social Network Analysis," Thousand Oaks, California: SAGE Publications, Inc., 2020. [Online]. Available: <https://methods.sagepub.com/book/mono/social-network-analysis-3e/chpt/introduction-social-network-analysis>
- [33] S. A. Myers, C. Zhu, and J. Leskovec, "Information diffusion and external influence in networks," presented at the Proceedings of the 18th ACM SIGKDD international conference on Knowledge discovery and data mining, Beijing, China, 2012. [Online]. Available: <https://doi.org/10.1145/2339530.2339540>.
- [34] Z. Katona, P. P. Zubcsek, and M. Sarvary, "Network Effects and Personal Influences: The Diffusion of an Online Social Network," *Journal of Marketing Research*, vol. 48, no. 3, pp. 425-443, 2011, doi: 10.1509/jmkr.48.3.425.

- [35] M. Kim, I. Baek, and M. Song, "Topic diffusion analysis of a weighted citation network in biomedical literature," *Journal of the Association for Information Science and Technology*, vol. 69, no. 2, pp. 329-342, 2018, doi: <https://doi.org/10.1002/asi.23960>.
- [36] X. Liu, S. Jiang, H. Chen, C. A. Larson, and M. C. Roco, "Nanotechnology knowledge diffusion: measuring the impact of the research networking and a strategy for improvement," *Journal of Nanoparticle Research*, vol. 16, no. 9, p. 2613, 31 Aug. 2014, doi: 10.1007/s11051-014-2613-x.
- [37] S. Molaei, S. Babaei, M. Salehi, and M. Jalili, "Information Spread and Topic Diffusion in Heterogeneous Information Networks," *Scientific Reports*, vol. 8, no. 1, p. 9549, 22 Jun. 2018, doi: 10.1038/s41598-018-27385-2.
- [38] X. Gao and J. Guan, "Network model of knowledge diffusion," *Scientometrics*, vol. 90, no. 3, pp. 749-762, 2012, doi: 10.1007/s11192-011-0554-z.
- [39] E. Yan and Y. Ding, "Applying centrality measures to impact analysis: A coauthorship network analysis," *Journal of the American Society for Information Science and Technology*, vol. 60, no. 10, pp. 2107-2118, 2009, doi: <https://doi.org/10.1002/asi.21128>.
- [40] J. Ravenscroft, M. Liakata, A. Clare, and D. Duma, "Measuring scientific impact beyond academia: An assessment of existing impact metrics and proposed improvements," *PLOS ONE*, vol. 12, no. 3, p. e0173152, 2017, doi: 10.1371/journal.pone.0173152.
- [41] M. E. J. Newman, "A measure of betweenness centrality based on random walks," *Social Networks*, vol. 27, no. 1, pp. 39-54, 2005, doi: <https://doi.org/10.1016/j.socnet.2004.11.009>.
- [42] S. Brin and L. Page, "The anatomy of a large-scale hypertextual Web search engine," *Computer Networks and ISDN Systems*, vol. 30, no. 1, pp. 107-117, 1 Apr. 1998, doi: [https://doi.org/10.1016/S0169-7552\(98\)00110-X](https://doi.org/10.1016/S0169-7552(98)00110-X).
- [43] Z. Gyöngyi, H. Garcia-Molina, and J. Pedersen, "Combating web spam with trustrank," presented at the Proceedings of the Thirtieth international conference on Very large data bases - Volume 30, Toronto, Canada, 2004.
- [44] L. Lü, D. Chen, X.-L. Ren, Q.-M. Zhang, Y.-C. Zhang, and T. Zhou, "Vital nodes identification in complex networks," *Physics Reports*, vol. 650, pp. 1-63, 13 Sep. 2016, doi: <https://doi.org/10.1016/j.physrep.2016.06.007>.
- [45] M. Dunaiski, W. Visser, and J. Geldenhuys, "Evaluating paper and author ranking algorithms using impact and contribution awards," *Journal of Informetrics*, vol. 10, no. 2, pp. 392-407, 1 May 2016, doi: <https://doi.org/10.1016/j.joi.2016.01.010>.
- [46] E. Yan, "Topic-based Pagerank: toward a topic-level scientific evaluation," *Scientometrics*, vol. 100, no. 2, pp. 407-437, 1 Aug. 2014, doi: 10.1007/s11192-014-1308-5.
- [47] F. Zhao, Y. Zhang, J. Lu, and O. Shai, "Measuring academic influence using heterogeneous author-citation networks," *Scientometrics*, vol. 118, no. 3, pp. 1119-1140, 1 Mar. 2019, doi: 10.1007/s11192-019-03010-5.
- [48] F. A. Massucci and D. Docampo, "Measuring the academic reputation through citation networks via PageRank," *Journal of Informetrics*, vol. 13, no. 1, pp. 185-201, 1 Feb. 2019, doi: <https://doi.org/10.1016/j.joi.2018.12.001>.
- [49] V. Freschi, "Protein function prediction from interaction networks using a random walk ranking algorithm," in *2007 IEEE 7th International Symposium on BioInformatics and BioEngineering*, 14-17 Oct. 2007, pp. 42-48, doi: 10.1109/BIBE.2007.4375543.
- [50] B. L. Mooney, L. R. Corrales, and A. E. Clark, "MolecularNetworks: An integrated graph theoretic and data mining tool to explore solvent organization in molecular simulation," *Journal of Computational Chemistry*, vol. 33, no. 8, pp. 853-860, 2012, doi: <https://doi.org/10.1002/jcc.22917>.

- [51] Z. Li, D. Fu, and J. He, "Everything Evolves in Personalized PageRank," presented at the Proceedings of the ACM Web Conference 2023, Austin, TX, USA, 2023. [Online]. Available: <https://doi.org/10.1145/3543507.3583474>.
- [52] T. H. Haveliwala, "Topic-sensitive PageRank," presented at the Proceedings of the 11th international conference on World Wide Web, Honolulu, Hawaii, USA, 2002. [Online]. Available: <https://doi.org/10.1145/511446.511513>.
- [53] D. Walker, H. Xie, K.-K. Yan, and S. Maslov, "Ranking scientific publications using a model of network traffic," *Journal of Statistical Mechanics: Theory and Experiment*, vol. 2007, no. 06, p. P06010, 14 Jun. 2007, doi: 10.1088/1742-5468/2007/06/P06010.
- [54] W. Xing and A. Ghorbani, "Weighted PageRank algorithm," in *Proceedings. Second Annual Conference on Communication Networks and Services Research, 2004.*, 21 May 2004, pp. 305-314, doi: 10.1109/DNSR.2004.1344743.
- [55] B. Chang, T. Xu, Q. Liu, and E.-H. Chen, "Study on Information Diffusion Analysis in Social Networks and Its Applications," *International Journal of Automation and Computing*, vol. 15, no. 4, pp. 377-401, 1 Aug. 2018, doi: 10.1007/s11633-018-1124-0.
- [56] D. Kempe, J. Kleinberg, and É. Tardos, "Maximizing the spread of influence through a social network," presented at the Proceedings of the ninth ACM SIGKDD international conference on Knowledge discovery and data mining, Washington, D.C., 2003. [Online]. Available: <https://doi.org/10.1145/956750.956769>.
- [57] T. Zhu, B. Wang, B. Wu, and C. Zhu, "Maximizing the spread of influence ranking in social networks," *Information Sciences*, vol. 278, pp. 535-544, 10 Sep. 2014, doi: <https://doi.org/10.1016/j.ins.2014.03.070>.
- [58] R. Zafarani, M. A. Abbasi, and H. Liu, *Social Media Mining: An Introduction*. Cambridge: Cambridge University Press, 2014.
- [59] D. Trpevski, W. K. S. Tang, and L. Kocarev, "Model for rumor spreading over networks," *Physical Review E*, vol. 81, no. 5, p. 056102, 05 Apr. 2010, doi: 10.1103/PhysRevE.81.056102.
- [60] L. M. A. Bettencourt, D. I. Kaiser, J. Kaur, C. Castillo-Chávez, and D. E. Wojick, "Population modeling of the emergence and development of scientific fields," *Scientometrics*, vol. 75, no. 3, pp. 495-518, 1 Jun. 2008, doi: 10.1007/s11192-007-1888-4.
- [61] K. Saito, M. Kimura, K. Ohara, and H. Motoda, "Efficient discovery of influential nodes for SIS models in social networks," *Knowledge and Information Systems*, vol. 30, no. 3, pp. 613-635, 1 Mar. 2012, doi: 10.1007/s10115-011-0396-2.
- [62] J. E. Hirsch, "An index to quantify an individual's scientific research output," *Proceedings of the National Academy of Sciences*, vol. 102, no. 46, pp. 16569-16572, 2005, doi: 10.1073/pnas.0507655102.
- [63] "Web of Science: h-index information." Clarivate. Accessed: 28 August 2025 [Online]. Available: https://support.clarivate.com/ScientificandAcademicResearch/s/article/Web-of-Science-h-index-information?language=en_US.
- [64] L. Egghe, "Theory and practise of the g-index," *Scientometrics*, vol. 69, no. 1, pp. 131-152, 1 Oct. 2006, doi: 10.1007/s11192-006-0144-7.
- [65] "Measuring your research impact: G-Index." Cornell University Library. Accessed: 28 August 2025 [Online]. Available: <https://guides.library.cornell.edu/impact/author-impact-g>.
- [66] "The Clarivate Impact Factor." Clarivate. Accessed: 28 August 2025 [Online]. Available: https://clarivate.com/academia-government/essays/impact-factor/?utm_medium=Referral&utm_source=en-wikipedia-org.

- [67] W. Zhou, J. Gu, and Y. Jia, "h-Index-based link prediction methods in citation network," *Scientometrics*, vol. 117, no. 1, pp. 381-390, 1 Oct. 2018, doi: 10.1007/s11192-018-2867-7.
- [68] Q. Zhao and X. Feng, "Utilizing citation network structure to predict paper citation counts: A Deep learning approach," *Journal of Informetrics*, vol. 16, no. 1, p. 101235, 1 Feb. 2022, doi: <https://doi.org/10.1016/j.joi.2021.101235>.
- [69] C. McCarty, J. W. Jawitz, A. Hopkins, and A. Goldman, "Predicting author h-index using characteristics of the co-author network," *Scientometrics*, vol. 96, no. 2, pp. 467-483, 1 Aug. 2013, doi: 10.1007/s11192-012-0933-0.
- [70] M. Erdt, A. Nagarajan, S.-C. J. Sin, and Y.-L. Theng, "Altmetrics: an analysis of the state-of-the-art in measuring research impact on social media," *Scientometrics*, vol. 109, no. 2, pp. 1117-1166, 1 Nov. 2016, doi: 10.1007/s11192-016-2077-0.
- [71] S. A. Elmore, "The Altmetric Attention Score: What Does It Mean and Why Should I Care?," *Toxicologic Pathology*, vol. 46, no. 3, pp. 252-255, 2018, doi: 10.1177/0192623318758294.
- [72] D. Tomasoni and A. J. S. Coats, "The Altmetric Attention Score: how science tries to meet social media," *European Journal of Heart Failure*, vol. 23, no. 5, pp. 693-697, 2021, doi: <https://doi.org/10.1002/ejhf.2136>.
- [73] "How are outputs scored?" Altmetric.com. Accessed: 27 August 2025 [Online]. Available: <https://help.altmetric.com/support/solutions/articles/6000232839-how-are-outputs-scored->.
- [74] S. Lee. "Citation Networks in Practice: Case Studies and Future Directions in Digital Rhetoric and Genre Studies." Accessed: 28 August 2025 [Online]. Available: <https://www.numberanalytics.com/blog/citation-networks-in-practice#:~:text=Altmetrics%2C%20or%20alternative%20metrics%2C%20refer,of%20research%20impact%20and%20dissemination.>
- [75] A. Abbasi, K. S. K. Chung, and L. Hossain, "Egocentric analysis of co-authorship network structure, position and performance," *Information Processing & Management*, vol. 48, no. 4, pp. 671-679, 1 Jul. 2012, doi: <https://doi.org/10.1016/j.ipm.2011.09.001>.
- [76] E. Finkel, "Engineered Mouse Virus Spurs Bioweapon Fears," *Science*, vol. 291, no. 5504, pp. 585-585, 2001, doi: 10.1126/science.291.5504.585.
- [77] G. Gronvall, "H5N1: A Case Study for Dual-Use Research " Council on Foreign Relations, 2013. [Online]. Available: https://cdn.cfr.org/sites/default/files/pdf/2013/05/WP_Dual_Use_Research.pdf
- [78] M. J. Selgelid and L. Weir, "Reflections on the Synthetic Production of Poliovirus," *Bulletin of the Atomic Scientists*, vol. 66, no. 3, pp. 1-9, 2010, doi: 10.2968/066003001.
- [79] R. D. Sleator, "Ferretting out the facts behind the H5N1 controversy," *Bioengineered*, vol. 3, no. 3, pp. 141-145, 1 May 2012, doi: 10.4161/bbug.19943.
- [80] K. P. Saalbach, "Chapter Three - Gain-of-function research," in *Advances in Applied Microbiology*, vol. 120, G. M. Gadd and S. Sariaslani Eds.: Academic Press, 2022, pp. 79-111.
- [81] "Dual Use Research of Concern: Fouchier H5N1 Study Finally Published." European Centre for Disease Prevention and Control. Accessed: 28 August 2025 [Online]. Available: <https://www.ecdc.europa.eu/en/news-events/controversial-research-modifying-ah5n1-influenza-viruses-laboratory>.
- [82] K. Kupferschmidt. (2018, 19 Jan.) A paper showing how to make a smallpox cousin just got published. Critics wonder why. *Science*. Available: <https://www.science.org/content/article/paper-showing-how-make-smallpox-cousin-just-got-published-critics-wonder-why>

- [83] D. DiEuliis, K. Berger, and G. Gronvall, "Biosecurity Implications for the Synthesis of Horsepox, an Orthopoxvirus," *Health Security*, vol. 15, no. 6, pp. 629-637, 1 Dec. 2017, doi: 10.1089/hs.2017.0081.
- [84] V. Arnaboldi, M. Conti, M. La Gala, A. Passarella, and F. Pezzoni, "Ego network structure in online social networks and its impact on information diffusion," *Computer Communications*, vol. 76, pp. 26-41, 15 Feb. 2016, doi: <https://doi.org/10.1016/j.comcom.2015.09.028>.
- [85] D. Knoke and S. Yang, "Advanced Methods for Analyzing Networks," Thousand Oaks, California: SAGE Publications, Inc., 2020. [Online]. Available: <https://methods.sagepub.com/book/mono/social-network-analysis-3e/chpt/advanced-methods-analyzing-networks>
- [86] Y. Ding, "Scientific collaboration and endorsement: Network analysis of coauthorship and citation networks," *Journal of Informetrics*, vol. 5, no. 1, pp. 187-203, 2011, doi: <https://doi.org/10.1016/j.joi.2010.10.008>.
- [87] N. A. Christakis and J. H. Fowler, "Social contagion theory: examining dynamic social networks and human behavior," *Statistics in Medicine*, vol. 32, no. 4, pp. 556-577, 2013, doi: <https://doi.org/10.1002/sim.5408>.
- [88] "Altmetric Attention Score modifiers." Altmetric.com. Accessed: 26 August 2025 [Online]. Available: <https://help.altmetric.com/support/solutions/articles/6000234288-altmetric-attention-score-modifiers>.
- [89] P. Habibzadeh, "Decay of References to Web sites in Articles Published in General Medical Journals: Mainstream vs Small Journals," (in En), *Appl Clin Inform*, vol. 04, no. 04, pp. 455-464, 19 Dec. 2013, doi: 10.4338/ACI-2013-07-RA-0055.
- [90] D. H.-L. Goh and P. K. Ng, "Link decay in leading information science journals," *Journal of the American Society for Information Science and Technology*, vol. 58, no. 1, pp. 15-24, 2007, doi: <https://doi.org/10.1002/asi.20513>.
- [91] W. Koehler, "A longitudinal study of Web pages continued: a consideration of document persistence," *Information Research*, vol. 9, no. 2, pp. 9-2, 2004.
- [92] J. Priem, H. Piwowar, and R. Orr, "OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts," *arXiv preprint arXiv:2205.01833*, 2022.
- [93] OpenAlex. OpenAlex technical documentation: API Overview [Online] Available: <https://docs.openalex.org/how-to-use-the-api/api-overview>
- [94] PyAlex. (2025). [Online]. Available: <https://github.com/J535D165/pyalex>
- [95] "Welcome to Altmetric's Details Page API documentation!" Altmetric LLC. Accessed: 3 September 2025 [Online]. Available: <https://details-page-api-docs.altmetric.com/index.html#welcome-to-altmetric-s-details-page-api-documentation>.
- [96] PyAltmetric. (2024). Center for Open Science. [Online]. Available: <https://github.com/CenterForOpenScience/PyAltmetric>
- [97] Overton. Documentation for Overton's REST API [Online] Available: <https://app.overton.io/swagger.php>
- [98] Langdetect. (2021). Michal Danilák. [Online]. Available: <https://pypi.org/project/langdetect/>
- [99] Welcome to graph-tool's documentation! (2025). Tiago P. Peixoto. [Online]. Available: <https://graph-tool.skewed.de/static/docs/stable/>
- [100] T. P. Peixoto. "Graph-tool performance comparison." Accessed: 4 September 2025 [Online]. Available: <https://graph-tool.skewed.de/performance.html>.
- [101] SciPy. "scipy.stats.spearmanr." Accessed: 28 August 2025 [Online]. Available: <https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.spearmanr.html>.

- [102] "Average Percentile." InCites Benchmarking & Analytics. Accessed: 28 August 2025 [Online]. Available: <https://incites.zendesk.com/hc/en-gb/articles/25087394395153-Average-Percentile>.
- [103] "Mann-Whitney U-Test." DATAtab. Accessed: 28 August 2025 [Online]. Available: <https://datatab.net/tutorial/mann-whitney-u-test>.
- [104] SciPy. "scipy.stats.mannwhitneyu." Accessed: 28 August 2025 [Online]. Available: <https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.mannwhitneyu.html>.
- [105] S. I. Bangdiwala, "Understanding significance and p-values," *Nepal Journal of Epidemiology*, vol. 6, no. 1, pp. 522-24, 31 Mar. 2016, doi: 10.3126/nje.v6i1.14732.
- [106] E. Gómez-de-Mariscal et al., "Use of the p-values as a size-dependent function to address practical differences when analyzing large datasets," *Scientific Reports*, vol. 11, no. 1, p. 20942, 22 Oct. 2021, doi: 10.1038/s41598-021-00199-5.
- [107] S. S. Mangiafico, "Mann-Whitney U Test and Wilcoxon Rank-sum Test for Two-Sample Data," in *Summary and Analysis of Extension Program Evaluation in R*, vol. version 1.23.0, 2025.
- [108] M. Kane and S. Y. Esther, "Using Cliff's Delta as a Non-Parametric Effect Size Measure: An Accessible Web App and R Tutorial," *Practical Assessment, Research, and Evaluation*, vol. 29, no. 1, 22 Jan. 2024, doi: 10.7275/pare.1977.
- [109] A. Vargha and H. D. Delaney, "A Critique and Improvement of the CL Common Language Effect Size Statistics of McGraw and Wong," *Journal of Educational and Behavioral Statistics*, vol. 25, no. 2, pp. 101-132, 2000, doi: 10.3102/10769986025002101.
- [110] S. S. Mangiafico. "Vargha and Delaney's A." Accessed: 29 August 2025 [Online]. Available: <https://search.r-project.org/CRAN/refmans/rcompanion/html/vda.html>.
- [111] J. Goedhart, "Calculation of a distribution free estimate of effect size and confidence intervals using VBA/Excel," *bioRxiv*, p. 073999, 2016, doi: 10.1101/073999.
- [112] S. S. Mangiafico. "Cliff's delta." Accessed: 29 August 2025 [Online]. Available: <https://search.r-project.org/CRAN/refmans/rcompanion/html/cliffDelta.html>.
- [113] SciPy. "scipy.stats.pearsonr." Accessed: 29 August 2025 [Online]. Available: <https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.pearsonr.html>.

Appendix A – Node type scores

Source Type	Score	Justification
news_website	8	Typically associated with wide reach and mainstream visibility, news websites provide direct exposure to non-academic audiences via the internet and serve as a major amplifier of public attention.
blog_post	5	Blogs can influence both specialist and lay audiences, and are often indexed by search engines, making them a potential entry point for diverse readers.
wikipedia, video	5	Publicly accessible platforms with high levels of web traffic. Wikipedia especially is frequently consulted for background information and thus capable of shaping broad audience perceptions.
policy_document	3	Authoritative outputs targeted toward policymakers and institutional decision-makers, conferring influence despite narrower audience reach.
clinical_guidance	3	Comparable in authority and influence to policy documents, particularly within healthcare and medical practice contexts.
book, book_chapter, report	2	Long-form outputs with slower circulation and lower immediacy than online sources.
transcript	1	Moderately visible depending on the platform and subject matter; influence varies with accessibility and content relevance.
patent	1	Primarily relevant to technical or commercial stakeholders; limited visibility outside specialized audiences.
academic_article	1	Central to scholarly communication but generally characterized by low direct public visibility; impact is mediated through citation within academic literature.
working_paper	1	Pre-publication scholarly outputs with restricted reach, often read mainly within specialist research communities.

Source Type	Score	Justification
preprint	1	Functionally similar to working papers, though increasingly prominent in some fields; typically more visible to researchers than to the wider public.
editorial, review, letter	1	Provide scholarly commentary and synthesis; generally limited in public reach but can draw academic attention to specific debates or works.
peer_review	1	Internal to scholarly publishing processes and of niche visibility, engaging primarily with academic communities.
guideline	1	Serves as a default classification where more specific clinical or policy designations are absent; moderate influence depending on application context.
dataset	0.5	Useful as research infrastructure or in business contexts, but generally low visibility outside specialist domains.
dissertation	0.5	Unpublished academic output with very limited circulation; rarely read beyond the awarding institution or specialist researchers.
grant	0.5	Administrative or funding-related outputs, typically of narrow visibility and often not made publicly accessible.
paratext, periodical, reference_entry	0.25	Ancillary or supporting materials for scholarly work; little direct impact on wider attention flows.
retraction, erratum	0.25	Primarily of interest to meta-researchers and academic quality assurance processes; limited broader attention.
other	0.01	Category for sources of uncertain or unknown reach, treated as negligible in terms of attention potential.

Appendix B – Information hazard-related terms

Critical terms (score +8):

"dual use", "gain of function", "information hazard", "knowledge hazard",
"malicious use", "malicious actor", "misuse", "weaponization",
"bioterrorism", "engineered pandemic", "catastrophic risk", "existential risk",
"biological weapon", "biological warfare", "bioterrorist", "biohacker",
"responsible use", "biotechnology risk", "biological attack", "terrorism",
"counter-terrorism", "weapons of mass destruction", "military application"

High relevance terms (score +5):

"biosecurity threat", "export controls", "import controls", "military",
"biodefense", "biological hazard", "information security",
"information governance", "synthetic biology", "biological risk",
"biological threat", "bioethics", "arms control"

Medium Relevance terms (score +2):

"risk assessment", "security implications", "chemical weapon", "nuclear weapon",
"governance", "national security", "artificial intelligence", "chemical warfare",
"proliferation", "dilemma", "disastrous", "outrage", "disarmament", "debate",
"risk management", "risk analysis", "high-risk", "controversy", "controversial",
"governance framework", "weapon", "artificial general intelligence", "ai safety",
"geopolitical", "man-made", "chemical threat", "nuclear threat", "radiological threat",
"international security", "weapon", "adversarial", "proliferation", "warfare",
"non-proliferation", "war", "malicious", "CBRN", "battlefield", "bwc", "cwc",
"biotechnology", "biosecurity", "biosafety"

Other broad terms that may be of relevance (score +1):

"ethical", "ethics", "risk", "regulation", "regulatory", "safety", "defense", "disaster"

Appendix C – Experiment 1 results

Table 2: Experiment 1a – Impact of edge weights modification on model performance (baseline = UNW+MBV))

Alpha	Variant	Beta	Spearman ρ	Overlap top_100 (%)	Overlap top_500 (%)	Overlap top 1pct (%)	Overlap top 5pct (%)	Top_100 IH_pct (%)	Δ top_100 IH_pct (%)	Top_500 IH_pct (%)	Top 1pct_IH (%)	Top 5pct_IH (%)	Average IH rank percentile (AIHRP)	Δ AIHRP
0.85	UNW	n/a	1	100	100	100	100	25	23.6	23.46	13.48	0	43.98	0
0.85	BW	n/a	1	96	97	97.73	97.49	24	23.8	23.24	13.31	-1	44.12	0.14
0.85	BW_D	n/a	1	96	97	97.73	97.49	24	23.8	23.24	13.31	-1	44.12	0.14
0.85	BW_D_L	n/a	1	91	93.6	95.57	95.98	23	24.8	23.14	13.5	-2	43.84	-0.14
0.85	BW_D_L_T	n/a	0.99	89	93.4	93.95	94.82	23	24.2	22.7	13.39	-2	43.93	-0.06
0.85	EXP_D	3	0.99	90	92.2	93.95	92.55	21	23.6	22.92	12.85	-4	44.32	0.33
0.85	EXP_D_L	3	0.99	83	90	92	91.94	23	23	22.81	13.26	-2	43.65	-0.33
0.85	EXP_D_L_T	3	0.99	83	88.8	91.03	91.34	22	23	22.27	13.37	-3	43.67	-0.32
0.85	EXP_D	5	0.99	87	90.6	91.78	90.78	21	24	22.7	12.77	-4	44.13	0.15
0.85	EXP_D_L	5	0.99	82	88	90.7	90.71	23	22.8	22.38	13.29	-2	43.58	-0.4
0.85	EXP_D_L_T	5	0.99	81	88	90.49	90.34	20	23.2	22.05	13.24	-5	43.61	-0.37
0.85	EXP_D	7	0.99	85	89.4	91.03	90.39	19	24	22.59	12.81	-6	44.01	0.02
0.85	EXP_D_L	7	0.99	81	87.8	90.05	90.06	23	23	22.05	13.33	-2	43.56	-0.43
0.85	EXP_D_L_T	7	0.99	81	87.8	89.95	89.93	20	23.4	21.95	13.18	-5	43.59	-0.4
0.85	EXP_D	9	0.99	85	88.8	90.27	89.8	19	23.8	22.49	12.77	-6	43.93	-0.05
0.85	EXP_D_L	9	0.99	80	87.4	89.51	89.67	22	23	21.95	13.33	-3	43.54	-0.44
0.85	EXP_D_L_T	9	0.99	81	87.2	89.51	89.5	20	23	21.95	13.22	-5	43.57	-0.41
0.99	UNW	n/a	1	100	100	100	100	23	24	23.14	13.31	0	44.06	0
0.99	BW	n/a	0.99	95	97.4	97.73	97.23	22	24.4	23.14	13.16	-1	44.21	0.15
0.99	BW_D	n/a	0.99	95	97.4	97.73	97.23	22	24.4	23.14	13.16	-1	44.21	0.15
0.99	BW_D_L	n/a	0.99	89	94.6	94.05	95.64	24	24	22.81	13.42	1	43.9	-0.16
0.99	BW_D_L_T	n/a	0.99	87	93.8	93.3	94.34	22	23.8	22.7	13.24	-1	43.99	-0.07

Alpha	Variant	Beta	Spearman ρ	Overlap top_100 (%)	Overlap top_500 (%)	Overlap top 1pct (%)	Overlap top 5pct (%)	Top_100 IH_pct (%)	Δ top_100 IH_pct (%)	Top_500 IH_pct (%)	Top 1pct_IH (%)	Top 5pct_IH (%)	Average IH rank percentile (AIHRP)	Δ AIHRP
0.99	EXP_D	3	0.98	88	91.6	92.54	91.99	22	23.2	22.59	12.62	-1	44.42	0.36
0.99	EXP_D_L	3	0.99	80	89.4	89.62	91.21	23	22.6	22.49	13.09	0	43.68	-0.38
0.99	EXP_D_L_T	3	0.99	80	87.8	88.54	90.39	22	22.6	22.16	13.07	-1	43.7	-0.36
0.99	EXP_D	5	0.98	84	89.8	89.95	90.28	19	23.2	22.49	12.66	-4	44.21	0.15
0.99	EXP_D_L	5	0.99	77	87.8	87.46	89.74	22	22.4	22.16	13.03	-1	43.61	-0.45
0.99	EXP_D_L_T	5	0.99	77	86.8	87.35	89.26	21	22.4	21.95	12.92	-2	43.64	-0.42
0.99	EXP_D	7	0.99	82	88.6	88.97	89.5	20	23	22.38	12.57	-3	44.07	0.01
0.99	EXP_D_L	7	0.99	76	87	86.38	89.03	22	22.6	21.84	12.98	-1	43.58	-0.48
0.99	EXP_D_L_T	7	0.99	76	86.8	86.59	88.85	21	22.4	21.84	12.9	-2	43.61	-0.45
0.99	EXP_D	9	0.99	80	87.8	87.78	88.81	19	22.6	22.38	12.59	-4	43.98	-0.08
0.99	EXP_D_L	9	0.99	76	86.4	86.27	88.57	22	22.8	21.84	13.05	-1	43.56	-0.5
0.99	EXP_D_L_T	9	0.99	74	86.6	86.49	88.42	20	22.4	21.84	12.92	-3	43.59	-0.47
0.5	UNW	n/a	1	100	100	100	100	25	23.4	24	13.91	0	43.76	0
0.5	BW	n/a	1	99	97.6	98.05	97.97	25	23.6	23.89	13.76	0	43.86	0.1
0.5	BW_D	n/a	1	99	97.6	98.05	97.97	25	23.6	23.89	13.76	0	43.86	0.1
0.5	BW_D_L	n/a	1	96	96.8	96.76	96.91	25	23.8	23.57	13.83	0	43.67	-0.08
0.5	BW_D_L_T	n/a	1	96	96	96	96.11	25	23.6	23.68	13.65	0	43.75	-0.01
0.5	EXP_D	3	0.99	98	95.6	95.68	94.92	26	23.4	23.57	13.63	1	44.03	0.27
0.5	EXP_D_L	3	1	94	94	94.16	94.1	26	23.4	23.35	13.7	1	43.56	-0.2
0.5	EXP_D_L_T	3	1	91	93.2	93.73	93.58	27	23.4	23.14	13.67	2	43.58	-0.18
0.5	EXP_D	5	0.99	95	93.6	93.84	93.58	26	23.4	23.68	13.5	1	43.91	0.15
0.5	EXP_D_L	5	1	91	92.8	93.3	93.32	26	23.2	23.46	13.63	1	43.52	-0.23
0.5	EXP_D_L_T	5	1	90	92.2	92.97	92.83	26	23.2	23.14	13.59	1	43.54	-0.21
0.5	EXP_D	7	0.99	95	91.4	93.3	93.11	26	23.4	23.57	13.48	1	43.82	0.06
0.5	EXP_D_L	7	1	90	91.4	92.97	92.89	26	23.4	23.35	13.65	1	43.51	-0.25
0.5	EXP_D_L_T	7	1	90	91.2	92.76	92.63	26	23.4	23.14	13.55	1	43.53	-0.23

Alpha	Variant	Beta	Spearman ρ	Overlap top_100 (%)	Overlap top_500 (%)	Overlap top 1pct (%)	Overlap top 5pct (%)	Top_100 IH_pct (%)	Δ top_100 IH_pct (%)	Top_500 IH_pct (%)	Top 1pct_IH (%)	Top 5pct_IH (%)	Average IH rank percentile (AIHRP)	Δ AIHRP
0.5	EXP_D	9	0.99	94	91.4	92.86	92.76	25	23.6	23.46	13.52	0	43.77	0.01
0.5	EXP_D_L	9	1	89	91.2	92.54	92.63	26	23.4	23.35	13.61	1	43.5	-0.26
0.5	EXP_D_L_T	9	1	89	90.6	92.65	92.31	25	23.2	23.14	13.55	0	43.52	-0.23

Table 3: Experiment 1b - sensitivity analysis for α

α	Variant	Beta	VS	Spearman's ρ	Overlap top_100 (%)	Overlap top_500 (%)	Overlap top 1pct (%)	Overlap top 5pct (%)	Top_100 IH_pct (%)	Δ top_100 IH_pct (%)	Top_500 IH_pct (%)	Top 1pct_IH (%)	Top 5pct_IH (%)	Average IH percentile rank (AIHPR)	Δ AIHRP
0.85	UNW	n/a	UNW @ $\alpha=0.85$	1	100	100	100	100	25	23.6	23.46	13.48	0	43.98	0
0.85	BW	n/a	BW @ $\alpha=0.85$	1	100	100	100	100	24	23.8	23.24	13.31	0	44.12	0
0.85	BW_D	n/a	BW_D @ $\alpha=0.85$	1	100	100	100	100	24	23.8	23.24	13.31	0	44.12	0
0.85	BW_D_L	n/a	BW_D_L @ $\alpha=0.85$	1	100	100	100	100	23	24.8	23.14	13.5	0	43.84	0
0.85	BW_D_L_T	n/a	BW_D_L_T @ $\alpha=0.85$	1	100	100	100	100	23	24.2	22.7	13.39	0	43.93	0
0.85	EXP_D	3	EXP_D @ $\alpha=0.85$	1	100	100	100	100	21	23.6	22.92	12.85	0	44.32	0
0.85	EXP_D_L	3	EXP_D_L @ $\alpha=0.85$	1	100	100	100	100	23	23	22.81	13.26	0	43.65	0
0.85	EXP_D_L_T	3	EXP_D_L_T @ $\alpha=0.85$	1	100	100	100	100	22	23	22.27	13.37	0	43.67	0
0.85	EXP_D	5	EXP_D @ $\alpha=0.85$	1	100	100	100	100	21	24	22.7	12.77	0	44.13	0
0.85	EXP_D_L	5	EXP_D_L @ $\alpha=0.85$	1	100	100	100	100	23	22.8	22.38	13.29	0	43.58	0
0.85	EXP_D_L_T	5	EXP_D_L_T @ $\alpha=0.85$	1	100	100	100	100	20	23.2	22.05	13.24	0	43.61	0
0.85	EXP_D	7	EXP_D @ $\alpha=0.85$	1	100	100	100	100	19	24	22.59	12.81	0	44.01	0

α	Variant	Beta	VS	Spearman's ρ	Overlap top_100 (%)	Overlap top_500 (%)	Overlap top 1pct (%)	Overlap top 5pct (%)	Top_100 IH_pct (%)	Δ top_100 IH_pct (%)	Top_500 IH_pct (%)	Top 1pct IH (%)	Top 5pct IH (%)	Average IH percentile rank (AIHPR)	Δ AIHRP
0.85	EXP_D_L	7	EXP_D_L @ $\alpha=0.85$	1	100	100	100	100	23	23	22.05	13.33	0	43.56	0
0.85	EXP_D_L_T	7	EXP_D_L_T @ $\alpha=0.85$	1	100	100	100	100	20	23.4	21.95	13.18	0	43.59	0
0.85	EXP_D	9	EXP_D @ $\alpha=0.85$	1	100	100	100	100	19	23.8	22.49	12.77	0	43.93	0
0.85	EXP_D_L	9	EXP_D_L @ $\alpha=0.85$	1	100	100	100	100	22	23	21.95	13.33	0	43.54	0
0.85	EXP_D_L_T	9	EXP_D_L_T @ $\alpha=0.85$	1	100	100	100	100	20	23	21.95	13.22	0	43.57	0
0.99	UNW	n/a	UNW @ $\alpha=0.85$	1	95	96	98.49	99.33	23	24	23.14	13.31	-2	44.06	0.08
0.99	BW	n/a	BW @ $\alpha=0.85$	1	95	96.2	98.81	99.07	22	24.4	23.14	13.16	-2	44.21	0.09
0.99	BW_D	n/a	BW_D @ $\alpha=0.85$	1	95	96.2	98.81	99.07	22	24.4	23.14	13.16	-2	44.21	0.09
0.99	BW_D_L	n/a	BW_D_L @ $\alpha=0.85$	1	98	98.2	97.62	99.24	24	24	22.81	13.42	1	43.9	0.06
0.99	BW_D_L_T	n/a	BW_D_L_T @ $\alpha=0.85$	1	97	98	97.95	99.07	22	23.8	22.7	13.24	-1	43.99	0.07
0.99	EXP_D	3	EXP_D @ $\alpha=0.85$	1	94	98.4	97.62	99.35	22	23.2	22.59	12.62	1	44.42	0.1
0.99	EXP_D_L	3	EXP_D_L @ $\alpha=0.85$	1	96	98.4	96.65	98.77	23	22.6	22.49	13.09	0	43.68	0.03
0.99	EXP_D_L_T	3	EXP_D_L_T @ $\alpha=0.85$	1	95	98	96.43	98.75	22	22.6	22.16	13.07	0	43.7	0.03

α	Variant	Beta	VS	Spear- man's ρ	Overlap top_100 (%)	Overlap top_500 (%)	Overlap top 1pct (%)	Overlap top 5pct (%)	Top_100 IH_pct (%)	Δ top_100 IH_pct (%)	Top_500 IH_pct (%)	Top 1pct_IH (%)	Top 5pct_IH (%)	Average IH percentile rank (AIHPR)	Δ AIHRP
0.99	EXP_D	5	EXP_D @ $\alpha=0.85$	1	90	98	96.76	99.33	19	23.2	22.49	12.66	-2	44.21	0.08
0.99	EXP_D_L	5	EXP_D_L @ $\alpha=0.85$	1	94	98.2	96	98.81	22	22.4	22.16	13.03	-1	43.61	0.02
0.99	EXP_D_L_T	5	EXP_D_L_T @ $\alpha=0.85$	1	95	97.6	96.32	98.64	21	22.4	21.95	12.92	1	43.64	0.02
0.99	EXP_D	7	EXP_D @ $\alpha=0.85$	1	90	98	96.86	98.98	20	23	22.38	12.57	1	44.07	0.06
0.99	EXP_D_L	7	EXP_D_L @ $\alpha=0.85$	1	94	97.8	95.68	98.62	22	22.6	21.84	12.98	-1	43.58	0.02
0.99	EXP_D_L_T	7	EXP_D_L_T @ $\alpha=0.85$	1	95	98	96.32	98.66	21	22.4	21.84	12.9	1	43.61	0.02
0.99	EXP_D	9	EXP_D @ $\alpha=0.85$	1	90	97.8	96.54	98.9	19	22.6	22.38	12.59	0	43.98	0.05
0.99	EXP_D_L	9	EXP_D_L @ $\alpha=0.85$	1	96	98	96.11	98.66	22	22.8	21.84	13.05	0	43.56	0.02
0.99	EXP_D_L_T	9	EXP_D_L_T @ $\alpha=0.85$	1	93	98.2	96.54	98.64	20	22.4	21.84	12.92	0	43.59	0.02
0.5	UNW	n/a	UNW @ $\alpha=0.85$	1	90	92.2	95.14	96.89	25	23.4	24	13.91	0	43.76	-0.23
0.5	BW	n/a	BW @ $\alpha=0.85$	1	88	93.6	94.81	97.47	25	23.6	23.89	13.76	1	43.86	-0.26
0.5	BW_D	n/a	BW_D @ $\alpha=0.85$	1	88	93.6	94.81	97.47	25	23.6	23.89	13.76	1	43.86	-0.26
0.5	BW_D_L	n/a	BW_D_L @ $\alpha=0.85$	1	90	91.6	95.68	97.26	25	23.8	23.57	13.83	2	43.67	-0.17

α	Variant	Beta	VS	Spearman's ρ	Overlap top_100 (%)	Overlap top_500 (%)	Overlap top 1pct (%)	Overlap top 5pct (%)	Top_100 IH_pct (%)	Δ top_100 IH_pct (%)	Top_500 IH_pct (%)	Top 1pct IH (%)	Top 5pct IH (%)	Average IH percentile rank (AIHPR)	Δ AIHRP
0.5	BW_D_L_T	n/a	BW_D_L_T @ $\alpha=0.85$	1	88	91.8	95.89	97.28	26	3	23.4	23.46	13.57	43.85	-0.22
0.5	EXP_D	3	EXP_D @ $\alpha=0.85$	1	85	90.4	94.38	96.63	26	5	23.4	23.57	13.59	44.09	-0.28
0.5	EXP_D_L	3	EXP_D_L @ $\alpha=0.85$	1	86	92.4	95.24	96.52	26	3	23.4	23.35	13.65	43.62	-0.09
0.5	EXP_D_L_T	3	EXP_D_L_T @ $\alpha=0.85$	1	89	91.6	95.03	96.65	27	5	23.4	23.14	13.63	43.64	-0.09
0.5	EXP_D	5	EXP_D @ $\alpha=0.85$	1	87	91.4	95.24	96.52	26	5	23.4	23.68	13.46	43.97	-0.22
0.5	EXP_D_L	5	EXP_D_L @ $\alpha=0.85$	1	88	92.2	94.81	96.87	26	3	23.2	23.46	13.59	43.58	-0.06
0.5	EXP_D_L_T	5	EXP_D_L_T @ $\alpha=0.85$	1	88	92.2	95.24	96.67	26	6	23.2	23.14	13.55	43.6	-0.07
0.5	EXP_D	7	EXP_D @ $\alpha=0.85$	1	86	92.4	95.14	96.78	26	7	23.4	23.57	13.44	43.88	-0.18
0.5	EXP_D_L	7	EXP_D_L @ $\alpha=0.85$	1	89	93.4	94.49	96.59	26	3	23.4	23.35	13.61	43.57	-0.05
0.5	EXP_D_L_T	7	EXP_D_L_T @ $\alpha=0.85$	1	88	93	94.59	96.41	26	6	23.4	23.14	13.5	43.59	-0.06
0.5	EXP_D	9	EXP_D @ $\alpha=0.85$	1	87	92.2	95.03	96.67	25	6	23.6	23.46	13.48	43.83	-0.16
0.5	EXP_D_L	9	EXP_D_L @ $\alpha=0.85$	1	89	92.8	94.27	96.54	26	4	23.4	23.35	13.57	43.56	-0.05
0.5	EXP_D_L_T	9	EXP_D_L_T @ $\alpha=0.85$	1	89	93	94.59	96.61	25	5	23.2	23.14	13.5	43.58	-0.05